Unit 2 Overview: Proofs and Logic

Unit Narrative

This unit is designed to build a student's understanding of the structures and thinking of proof. Teachers should allow students to struggle and grapple with their learning. Students who struggle may be better with a stronger partner but not one who will just do the work for them. Many lessons begin with collaboration or include an activity. Research shows that exploratory learning and making mistakes have a stronger impact on student retention. Teachers should circulate and support where necessary. Teachers could incorporate check-in points by having students have the teacher visit their pair at a certain point of the exploration or activity. Many of the warmups and wrap-ups are meant to just activate the brain. The content will be cognitively demanding. The end goal is that students are supported in understanding the role of a proof and the logical thinking that support this. The teacher could choose to utilize a formative assessment for this unit knowing that student knowledge of proofs will mature throughout the course. This is also a unit that the teacher can refer to when helping a student who is struggling with the flow of a proof. The last proof in lesson 8 is much more complex. This was done so that students see a proof that is not a direct flow of thinking. All other proofs are direct – all statements flow from the given whereas the last proof in lesson 8 must flow from two givens. This unit lays important groundwork and should be seen as such.

Section 1 begins by introducing students to Euclidean geometry to establish essential vocabulary and concepts such as postulates, theorems, and proofs. It will then transition students to the different types of reasoning, including conjectures, counterexamples, deductive reasoning, and inductive reasoning. They will then use this knowledge to determine logical conclusions. This section concludes with an introduction to the hypothesis and the conclusion of a statement. It includes converse, inverse, and contrapositive of a conditional statement.

Section 2 extends students' work with converse, inverse, and contrapositive of a conditional statement to the concept of a biconditional statement. They will then move on to "all" and "not" statements as they wrap up conditional and biconditional statements. Students will then apply this and everything else learned in the unit to formally move into proofs. Lessons 7 and 8 provide students with a two-lesson series on paragraph proofs, flowchart proofs, and two-column proofs.

This unit contains several new vocabulary terms. Consider leveraging vocabulary strategies such as a word wall, Frayer Models, anchor charts, guided notes, or other alternative strategies.

Unit At-A-Glance

Section 1: Logic and Reasoning	Benchmarks Addressed	Lesson 1	Introduction to Euclidean Geometry
	MA.912.GR.1.1 **MA.912.LT.4.10**	Lesson 2	Types of Reasoning
		Lesson 3	Logical Conclusions
Section 2: Conditional Statements	Benchmarks Addressed	Lesson 4	Hypothesis and Conclusion
	MA.912.LT.4.3	Lesson 5	Conditionals and Biconditionals
		Lesson 6	"All" and "Not" Statements
Section 3: Proofs	Benchmarks Addressed	Lesson 7	Proofs – Part 1
	MA.912.GR.1.1 **MA.912.LT.4.10**	Lesson 8	Proofs – Part 2

Differentiation

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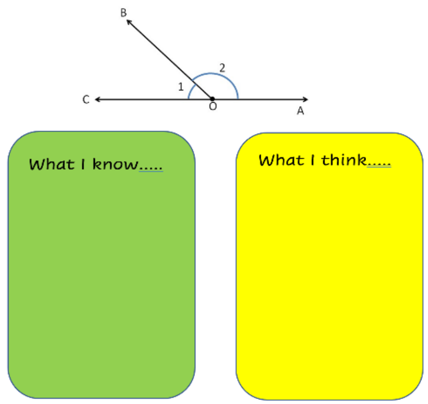
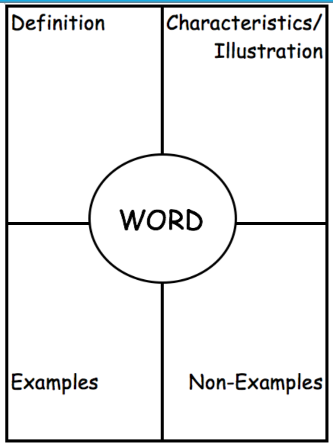
Engagement *The "WHY" of learning*	• Section 2: Conditional Statements offers an opportunity to engage students through relevant and interesting context. Consider having students bring in examples of advertisements that can be used to write conditional statements. Post images of the advertisement with the conditionals written for extra visuals. Use the advertisements throughout all of Section 2 to provide a creative and interesting way to make sense of conditional statements, converses, inverses, contrapositives, biconditionals, and "all" and "not" statements.  If you eat Subway, then you eat fresh.
Representation *The "WHAT" of learning*	• Vocabulary used in Geometry is building on vocabulary learned in previous courses. Students will enter the class with different exposure to this vocabulary due to their progression through middle school courses. Students will also enter with varying levels of retention of the previous vocabulary. Anchor instruction by linking to and activating relevant prior knowledge using: ○ visual imagery (pictures, anchor charts) ○ prerequisite checks (What I Know, What I Think) ○ pre-teaching • Instruction can then be targeted to address the entry point of each learner. 
Action & Expression *The "HOW" of learning*	• Did your students enjoy reading the quotes in Lesson 4? Challenge students to bring to class some of their favorite quotes written in "If, then" form or to write their own quotes for a fun project idea.

ELL Information



Section 2 is heavy on new terms that students can oftentimes get confused with one another. Throughout this section, be sure to use various language and vocabulary strategies to help students connect to the concept. Consider the use of Frayer Models, anchor charts, guided notes, or other strategies that may be effective with your students. Look for opportunities to connect the definitions of these terms to examples that students can relate to and that are appealing to the personalities of your students.

All glossary terms are available in multiple languages, including English, Spanish, Haitian Creole, and American Sign Language, as support for ELL students. Consider pairing the following videos with strategies from "Additional Differentiated Support" below:

- Acute angle
- Congruent
- Complementary angles
- Irrational number
- Polygon
- Prime number
- Quadrilateral
- Rational number
- Reflexive property of equality
- Substitution property of equality
- Symmetric property of equality
- Transitive property of equality
- Supplementary angles
- Vertical angles

Additional Differentiated Support

Scaffolding Resources	On-Ramp
Whenever appropriate, consider using physical manipulatives or physical representations to provide hands-on opportunities during lessons. During any proof work, adding or removing scaffolding by providing more or less completed statements and reasons will help to increase or decrease the cognitive demand of tasks based on student need.	Use the videos and practice problems in the On-Ramp to Geometry personalized learning tool to review the following topics for this unit. Videos are available in English and Spanish. • Parallel and Perpendicular Lines • Right, Acute, and Obtuse Angles

Section Coherence

Section 1: Logic and Reasoning (Lessons 1-3)

<table>
<tr><th>Benchmarks Addressed</th><th colspan="2">Learning Targets</th></tr>
<tr><td rowspan="3">MA.912.GR.1.1 Prove relationships and theorems about lines and angles. Solve mathematical and real-world problems involving postulates, relationships, and theorems of lines and angles.
• Clarification 1: Postulates, relationships, and theorems including vertical angles are congruent; when a transversal crosses parallel lines, the consecutive angles are supplementary and alternate (interior and exterior) angles and corresponding angles are congruent; points on a perpendicular bisector of a line segment are exactly those equidistant from the segment’s endpoints.
• Clarification 2: Instruction includes constructing two-column proofs, pictorial proofs, paragraph and narrative proofs, flowchart proofs or informal proofs.
• Clarification 3: Instruction focuses on helping a student choose a method they can use reliably.

MA.912.LT.4.10 Judge the validity of arguments and give counterexamples to disprove statements.
• Clarification 1: Within the Geometry course, instruction focuses on the connection to proofs within the course.</td><td>Lesson 1</td><td>• I can distinguish between definition, postulate, and theorem.
• I understand the role of proof.
• I can relate properties of equality to the properties of congruence.</td></tr>
<tr><td>Lesson 2</td><td>• I can write a conjecture.
• I can write a counterexample.
• I can use inductive and deductive reasoning.</td></tr>
<tr><td>Lesson 3</td><td>• I can complete an argument that leads to a logical conclusion.
• I can determine if an argument leads to a logical conclusion.</td></tr>
</table>

<table>
<tr><th colspan="3">Section 2: Conditional Statements (Lessons 4-6)</th></tr>
<tr><th>Benchmarks Addressed</th><th colspan="2">Learning Targets</th></tr>
<tr><td rowspan="3">MA.912.LT.4.3 Identify and accurately interpret “if…then,” “if and only if,” “all” and “not” statements. Find the converse, inverse and contrapositive of a statement.
Clarification 1: Instruction focuses on recognizing the relationships between an “if…then” statement and the converse, inverse and contrapositive of that statement.
Clarification 2: Within the Geometry course, instruction focuses on the connection to proofs within the course.</td><td>Lesson 4</td><td>I can identify the hypothesis of a statement.
I can identify the conclusion of a statement.
I can write the converse, inverse, and contrapositive of a conditional statement.</td></tr>
<tr><td>Lesson 5</td><td>I can write the converse, inverse, and contrapositive of a mathematical statement.
I can determine if a statement is biconditional.
I can write a definition as a biconditional.</td></tr>
<tr><td>Lesson 6</td><td>I can write and interpret an “all” statement.
I can write and interpret a negation.</td></tr>
</table>

<table>
<tr><th colspan="3">Section 3: Proofs (Lessons 7-8)</th></tr>
<tr><th>Benchmarks Addressed</th><th colspan="2">Learning Targets</th></tr>
<tr><td rowspan="2">MA.912.GR.1.1 Prove relationships and theorems about lines and angles. Solve mathematical and real-world problems involving postulates, relationships and theorems of lines and angles.
Clarification 1: Postulates, relationships and theorems including vertical angles are congruent; when a transversal crosses parallel lines, the consecutive angles are supplementary and alternate (interior and exterior) angles and corresponding angles are congruent; points on a perpendicular bisector of a line segment are exactly those equidistant from the segment’s endpoints.
Clarification 2: Instruction includes constructing two-column proofs, pictorial proofs, paragraph and narrative proofs, flowchart proofs or informal proofs.
Clarification 3: Instruction focuses on helping a student choose a method they can use reliably.
MA.912.LT.4.10 Judge the validity of arguments and give counterexamples to disprove statements.
Clarification 1: Within the Geometry course, instruction focuses on the connection to proofs within the course.</td><td>Lesson 7</td><td>I can relate deductive reasoning to completing proofs based on an assumption or given.
I can recognize paragraph, flowchart, and two column proofs.</td></tr>
<tr><td>Lesson 8</td><td>I can reflect on the mathematical thinking found in a proof.
I can reason about proofs using deductive reasoning, definitions, and postulates.</td></tr>
</table>

Vertical Alignment

Grade 8 Unit 13 Properties and Theorems of Angles	→	**Geometry Unit 2** **Proofs and Logic**

Prior Course(s)	Course Level	Future Course(s)
MA.8.GR.1.4 Solve mathematical problems involving the relationships between supplementary, complementary, vertical, or adjacent angles.	**MA.912.GR.1.1** Prove relationships and theorems about lines and angles. Solve mathematical and real-world problems involving postulates, relationships, and theorems of lines and angles. **MA.912.LT.4.10** Judge the validity of arguments and give counterexamples to disprove statements. **MA.912.LT.4.3** Identify and accurately interpret "if…then," "if and only if," "all," and "not" statements. Find the converse, inverse, and contrapositive of a statement.	

2.1 Introduction to Euclidean Geometry

Lesson Introduction

Benchmarks	MA.912.GR.1.1 Prove relationships and theorems about lines and angles. Solve mathematical and real-world problems involving postulates, relationships, and theorems of lines and angles. MA.912.LT.4.10 Judge the validity of arguments and give counterexamples to disprove statements.	Learning Targets	• I can distinguish between definition, postulate, and theorem. • I understand the role of proof. • I can relate properties of equality to the properties of congruence.

Benchmark Coherence	Building On	Working Toward
	N/A	N/A

Essential Questions	• What is the difference between a definition, a postulate, and a theorem? • What is the role of a proof in Geometry? • How are the properties of equality related to the properties of congruence?
Lesson Narrative	In this lesson, students will begin by exploring the concept of Euclidean geometry. They will be introduced to the new vocabulary “postulate” and “theorem.” Using the vertical angles theorem, students will observe how a pictorial representation, or proof, can be used to build toward more formal proofs. The lesson then moves students into formal connections of the reflexive, symmetric, and transitive properties as properties of equality and congruence. Distinction is made between the transitive property of equality and the substitution property of equality and also between equality and congruence.

Component Summary				
	2.1.1 Warm-Up (~5 min.)	Math with number cubes *(SE p. 83)*	2.1.3 Guided Instruction (15 min.)	Connections between properties of equality and congruence *(SE p. 86)*
	2.1.2 Exploration (25 min.)	Exploration into Euclidean geometry and proofs *(SE p. 84)*	2.1.4 Wrap-Up (~5 min.)	Completing a table with terms known, things wondered, and things learned *(SE p. 88)*

Resources	Glossary	Materials	Additional Preparation
	<u>Key Terms:</u> • Postulate • Theorem • Vertical Angles Theorem • Definition of congruence <u>Additional Terms:</u> • Reflexive • Symmetric • Transitive • Equality • Congruence	• (Optional) Patty paper • (Optional) Protractor • (Optional) Compass • (Optional) Ruler	• This lesson serves as the foundation of the unit and introduces essential vocabulary. Review the vocabulary and historical figures to have a better understanding of what is going to be presented to students. • If one has not already been started, consider having the students create a proof resource booklet, notebook, or type of study guide that will continue to be added to throughout the course. This proof reference would include definitions, postulates, and theorems.

<table>
<tr><td rowspan="4">Teacher Tips</td><th>Common Misconceptions</th><th>Things to Consider</th></tr>
<tr><td>

Students may mistake the concept of a postulate and a theorem.
Students may mix up an equality statement and a congruence statement.
Students may not recall the correct property for a given statement.

</td><td>

This lesson is heavy with introductory information and rich in vocabulary. Consider a way to capture the terms introduced in the lesson such as a guided notes sheet, foldable, or Frayer model.
Consider staggering 2.1.2 Exploration by having students stop when they get to a new vocabulary term (blue box) so it can be discussed as a whole group.
Consider creating manipulatives using card stock for 2.1.2 Exploration.

</td></tr>
<tr><th>Literacy and Language Routines</th><th>Mathematical Thinking and Reasoning (MTR) Standard</th></tr>
<tr><td>Notice and Wonder
2.1.1 Warm-Up provides students with an activity that shows a picture of 16 number cubes. Students will be asked about what they notice. Consider also asking students what they wonder to fully leverage a Notice and Wonder protocol.</td><td>MA.K12.MTR.4.1: Discussion and Reflection
This lesson is an introduction to Geometry and focused on the use of MA.K12.MTR.4.1. Through the 2.1.2 Exploration, students are engaged in discussions about mathematical ideas and vocabulary. Students justify results and construct possible arguments. Teachers should establish a culture in which students ask questions and understand mistakes as an opportunity to learn.</td></tr>
<tr><td>Differentiate Instruction</td><td colspan="2">Consider pairing students who struggle with a stronger partner who can maintain collaboration so one student is not doing the primary heavy lift of the Exploration in this lesson.</td></tr>
<tr><td colspan="3">Lesson Notes:</td></tr>
</table>

2.1.1 Warm-Up: Number Cubes Math

Component Overview: *Students will be presented with 16 number cubes and will respond to three questions about them. For questions 2 and 3, check that students are explaining their work, as this will highlight their growth toward stronger number sense.*

2.1.1 Warm-Up: Number Cubes Math

A picture of 16 number cubes is shown.

1. What do you notice?
   Answers will vary. I notice that half of the dice are blue with a number 4 on them, and the other half are white with a 3 on them.

2. Find the sum of the number cubes. Show your work or explain your thinking.
   Answers will vary. I counted the number of cubes with 3 dots and found that there are 8 of them. Then, I counted the number of cubes with 4 dots and found that there are 8 of them as well. I multiplied the number of dots with the number of cubes and added the products. $8 \cdot 3 = 24$ also $8 \cdot 4 = 32$ then to add $24 + 32 = 56$ so 56 is the sum of the number cubes.

3. If all of the number cubes with 3 dots were changed to number cubes with 6 dots, how would you find the new sum? Show your work or explain your thinking.
   Answers will vary. If the dots on all the number cubes with 3 dots were doubled, then we can double the sum we found previously as 24 to 48. Now I add the 48 to the same 32 that we got from all the cubes with a 4 to get $48 + 32 = 80$. So, the new sum would be 80.

If time permits, consider having students share any reasonable things that they notice about the image. Don't limit the positive responses to what students notice to things that are mathematically appropriate. For example, if a student says there are blue and white dice, that is a valid observation.

Consider capturing all the things students notice on chart paper or a whiteboard and provide multiple opportunities for student voice.

2.1.2 Exploration: What Is Euclidean Geometry?

Component Overview: *Students will explore key introductory vocabulary, postulates, and theorems of Euclidean geometry. Allow students to work in pairs to complete the Exploration.*

2.1.2 Exploration: What is Euclidean Geometry?

Euclidean geometry is the study of planes and solid figures on the basis of axioms and theorems compiled by the Greek mathematician Euclid. Rather than using algorithms to solve problems like you may have done in other math courses, Euclidean geometry emphasizes using facts, ideas, and assumptions.

Euclid is known as the father of Geometry. He wrote 13 books using an axiomatic system to construct bodies of knowledge.

Each of Euclid's 13 books have proofs. Each proof shows how a hypothesis is true. The proof uses definitions, postulates, and proven theorems to explain why the hypothesis is true.

A ***postulate*** is an idea that is assumed to be true or something that is self-evident.

The first postulate in Euclid's Book 1 is shown.

A straight line segment can be drawn joining any two points.

1. Use a straightedge to show the postulate using the points shown.

For the introductory paragraph, consider highlighting keywords that may be unfamiliar to students prior to releasing the students to read. This may provide students a better understanding of the information. For example, the words "axiom," "theorem," and "proven notions" may be unfamiliar.

This exploration is scaffolded to allow students to engage with it with their partner rather than being guided through by the teacher. Allow students the opportunity to revise their thinking throughout the exploration as they engage with each new reflection. Observe and listen as students work with their partners. Make note of student misconceptions that should be addressed. Additional scaffolding questions are provided, if needed, but this should be a primarily student-directed exploration.

Students will likely not be familiar with who Euclid was or his impact on geometry. Consider having students further explore the history of Geometry, Euclid, or Euclid's 13 books.

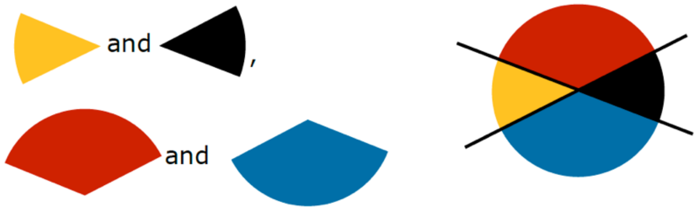
2.1.2 Exploration: What Is Euclidean Geometry? (continued)

2. Discuss with your partner why a postulate is something that is self-evident. Summarize your discussion.
   Answers will vary.

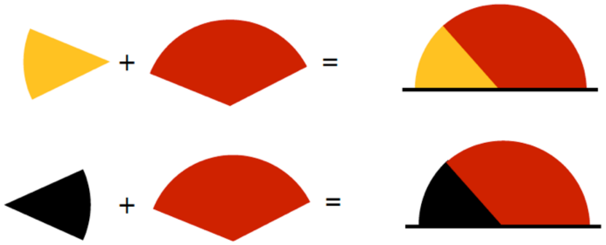
A ***theorem*** is an idea that has to be shown to be true through a chain of reasoning, often called a proof.

Vertical Angles Theorem: If two lines intersect, then vertical angles are congruent.

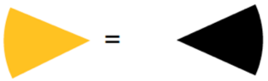
The vertical angles are



A pictorial proof of the vertical angles theorem is shown.



Therefore,

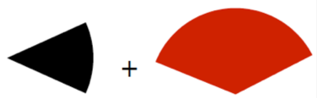
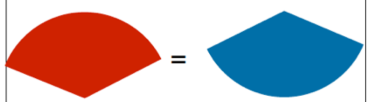


In the same manner,

Students may need assistance with organizing the new vocabulary of this lesson. Consider using an anchor chart, foldable, or Frayer Model for each of these terms.
When reviewing the new vocabulary, it is important for students to understand the similarities between them, but more importantly, what makes them different. For example, they should see how a theorem is not the same as a postulate and why.

Students are first introduced to vertical angles in benchmark MA.8.GR.1.4. Students may struggle with the pictorial proof. Consider having patty paper, protractors, and other tools available for students to explore the proof.
If students are creating a proof resource booklet, notebook, or type of study guide, add the vertical angles theorem.

3. Discuss with your partner the pictorial proof of the vertical angles theorem below. Complete the description for each picture shown in the pictorial proof.

Picture	Description
	The yellow angle and the red angle are being added.
	*The black angle and the red angle are being added.*
	*This image shows the yellow and red angles being added equal* $180°$.
	*The black angle added to the red angle equal* $180°$.
	*The yellow angle and the black angle are equal in size.*
	*The red angle and the blue angle are equal in size.*

Proofs play an important role in the course of geometry. Proofs can take on many forms. They are the set of logical steps that explain why something is true.

When programmers write a new game, they use a set of logical steps. When you tie your shoes, you use a set of logical steps.

If available, consider using physical manipulatives or representations here. The images provide a visual representation of the proof, but kinesthetic learners may still struggle with the activity. By allowing students to manipulate the shapes to match the pictures, it can help provide a hands-on opportunity during the Exploration.

Questions to consider:

- *What type of angle is formed by the angles in Image 3? (Straight angle.)*
- *How do you know that the third image shows two angles being added together add to 180º?*
- *How do you know that the yellow angle and the red angle add to be 180º?*
- *What is the relationship between the addition of the yellow and red angles and the addition of the black and red angles?*
- *How can you conclude that the yellow and black angles are equal in size?*
- *How do you know the red angle and the blue angles are equal in size?*

2.1.2 Exploration: What Is Euclidean Geometry? (continued)

4. Work with your partner to place the steps in tying a pair of shoes in order.

Step	Description	Picture
2	Make a loop in one lace.	
1	Cross the laces and pull them tight.	
4	Pull both loops tight.	
3	Wrap the other lace around the loop and push some lace through the hole that is made at the bottom to make a second loop.	

5. Discuss with your partner why someone who learned to tie their shoes using a different method may have difficulty putting these steps in a logical order. Summarize your discussion.
Answers will vary. If you learned to tie your shoes in a different way, you may use steps that are not in the table.

Consider spending a few moments discussing the benefits of proofs. Students will likely struggle as they begin working with proofs, and you may hear them articulate a wondering of why they need to learn them. Beyond mathematical applications, consider sharing additional benefits such as:

- *a means of communicating your reasoning with others.*
- *a justification for the way things work.*
- *a basis for developing other applications of logical reasoning.*

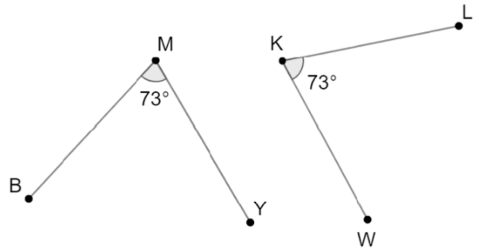
Consider pausing before moving forward to the next activity. Summarize the pictorial proof for the vertical angles theorem and the logical steps for tying a shoe.
Allow the Exploration to be free of the pressure of being correct but more of an Exploration of thinking.

2.1.3 Guided Instruction: Properties of Equality and Congruence

Component Overview: *Students will receive formal instructions on the properties of equality and congruence. Consider using gradual release during this activity.*

2.1.3 Guided Instruction: Properties of Equality and Congruence

In geometry, there are different notations for congruence and equality. For example, $\angle BMY$ and $\angle LKW$ have the same measure, so the angle measurements are equal. This is written as $m\angle BMY = m\angle LKW$. It can also be said that the angles are congruent, which is written as $\angle BMY \cong \angle LKW$.



Equality and congruence may seem to be the same, but equality is used when relationships can be expressed using numbers. Congruence is used when the relationship is not noted using numbers.

1. For each property, complete the example(s).

a. Reflexive property

Equality	Congruence
$m\angle WHD = m\angle\ WHD$	$\angle WHD \cong \angle WHD$

b. Symmetric property

Equality	Congruence
If $m\angle CKR = 59°$ then $59° = m\angle CKR$.	If $\angle CKR \cong \angle HDE$, then $\angle HDE \cong \angle CKR$.

c. Transitive property

Equality	Congruence
If $m\angle CRK = m\angle HWD$ and $m\angle HWD = 90°$, then $m\angle CRK = 90°$.	If $\angle CRK \cong \angle HWD$ and $\angle HWD \cong \angle MPX$, then $\angle CRK \cong \angle MPX$.

This component contains the importance of proper notation for equals and congruence. Pause to ask students to consider what is the same and/or different about the two statements $m\angle BMY = m\angle LKW$ and $\angle BMY \cong \angle LKW$.
While students may better understand the substitution property and its use as a correct justification, the transitive property will be used in this course where it is appropriate. Consider addressing the level of acceptance of either property in your class when appropriate.

2.1.3 Guided Instruction: Properties of Equality and Congruence (continued)

d. Substitution property

Equality
If $m\angle CRK + m\angle HWD = 156°$ and $m\angle CRK = 84°$, then $\underline{84° + m\angle HWD = 156°}$.

2. Write the conclusion for each statement. Then, name the property.

Statement	Property
If $m\angle FDT + m\angle BYH = 245°$ and $245° = m\angle MPW$, then $m\angle FDT + m\angle BYH = m\angle MPW$.	*Transitive Property of Equality*
If $m\angle FDT + m\angle BYH = m\angle RLC + m\angle KNW$ and $m\angle RLC = m\angle BYH$, then $m\angle FDT + m\angle BYH = m\angle BYH + m\angle KNW$.	*Substitution Property of Equality*

3. Discuss with your partner the conditional statement, "If $m\angle CRK = m\angle HWD$, then $\angle CRK \cong \angle HWD$." Summarize your discussion.
When the measures of two angles are equal, then those two angles must also be congruent.

In geometry, there is a formal difference between equality and congruence. Equality refers to measurements, which is why $m\angle CRK = m\angle HWD$ includes the m for measurement. On the other hand, congruence means the shape is the same, which can be shown using isometry. In other words, the figure coincides exactly when superimposed.

If two figures have equal measurements, then by the **definition of congruence** the figures are congruent. If congruence is shown through a series of transformations, then the **definition of congruence in terms of rigid motions** was applied.

As a possible scaffold, consider providing students with a word bank with options of properties for students to choose from to complete the table. If you do this, make sure there are reasonable answer choices and quality distractors that would require the student to know which one represents the given statement.

Students may need to be referred back to Unit 1 to recall isometry and rigid motions.

2.1.3 Guided Instruction: Properties of Equality and Congruence (continued)

The definition of congruence can be used to change from congruence to equality.

Statement	Reason
1. $\angle CRK \cong \angle HWD$	1. Given
2. $m\angle CRK = m\angle HWD$	2. Definition of congruence

Or, the definition of congruence can be used to change from equality to congruence.

Statement	Reason
1. $m\angle CRK = m\angle HWD$	1. Given
2. $\angle CRK \cong \angle HWD$	2. Definition of congruence

Gottfried Wilhelm Leibniz (1646-1716) introduced a symbol using one horizontal bar under the tilde for congruence in an unpublished manuscript of 1679. In 1824, Carl Brandan Mollweide (1774-1825) used the modern congruent symbol with two horizontal bars under a tilde in the book, *Euklid's Elemente.*

The definition of congruence can be used to state that "two figures are congruent if their measures are equal" and to state that "two figures are equal if they are congruent." Do you think all definitions can be used "in reverse" like this? (This question sets up future learning that definitions are biconditionals.)

The historical information is here as an added historical context to the lesson. Students should not be expected to memorize names or dates related to the development of the modern-day congruency symbol.

2.1.4 Wrap-Up: Cool Down

Component Overview: *Students will complete a table related to what they learned in the Exploration of this lesson.*

2.1.4 Wrap-Up: Cool Down

1. Complete the table.
   Answers will vary.

The terms and information used in the Exploration and Guided Instruction that I knew were…	Something I wondered during the Exploration or Guided Instruction was…	Something I learned during the Exploration or Guided Instruction was…

While this activity is open-ended, it can provide you with information on what students recall from this lesson. It can let you know what concepts and vocabulary terms stuck with students.

2.2 Types of Reasoning

Lesson Introduction

Benchmark	MA.912.LT.4.10 Judge the validity of arguments and give counterexamples to disprove statements.	Learning Targets	• I can write a conjecture. • I can write a counterexample. • I can use inductive and deductive reasoning.
Benchmark Coherence	Building On: MA.8.GR.1.4 Solve mathematical problems involving the relationships between supplementary, complementary, vertical or adjacent angles.	Working Toward: N/A	
Essential Questions	• How can I write a conjecture? • How can I use the information provided to write a counterexample? • What are the differences in using inductive and deductive reasoning?		
Lesson Narrative	In this lesson, students are introduced to different types of reasoning. Students will begin by exploring types of reasoning, while being introduced to the formal definitions of a conjecture, deductive reasoning, and inductive reasoning. They will discover how definitions, postulates, properties, and theorems can play a role in drawing an appropriate conclusion. Students will also be introduced to the concept of a counterexample.		

Component Summary			
2.2.1 Warm-Up (~5 min.)	Determining a pattern activity *(SE p. 89)*	2.2.3 Bring It Together (10 min)	Identifying types of reasoning and counterexamples *(SE p. 93)*
2.2.2 Exploration (30 min)	Exploring different types of reasoning *(SE p. 90)*	2.2.4 Wrap-Up (~5 min.)	Students summarize their learning *(SE p. 93)*

Resources	Glossary	Materials	Additional Preparation
	Key Terms: • Conjecture • Deductive Reasoning • Inductive Reasoning • Counterexample Additional Terms: N/A	N/A	• Students may need more examples of statements that represent inductive and deductive reasoning. Have additional examples prepared if these are needed.

<table>
<tr><td rowspan="4">Teacher Tips</td><td>Common Misconceptions</td><td>Things to Consider</td></tr>
<tr><td>Students may confuse deductive reasoning with inductive reasoning.
Students may be able to apply the concepts to real-world scenarios but not apply to mathematical concepts.
Students may inadvertently write a converse statement instead of a counterexample.</td><td>This lesson is language- and vocabulary-heavy. It may be helpful for students to get extra practice in both real-world examples and mathematical scenarios.</td></tr>
<tr><td>Literacy and Language Routines</td><td>Mathematical Thinking and Reasoning (MTR) Standard</td></tr>
<tr><td>Think Pair Share
In the 2.2.2 Exploration activity, students will work together to make conjectures. Students will have the opportunity, in questions 1 and 2, to make a conjecture on their own and then work with their partner to share their answers while summarizing their reasons.</td><td>MA.K12.MTR.4.1: Discussion and Reflection
2.2.2 Exploration provides students with an opportunity to work with a partner to engage in active discussion and reflection based on the prompts and instructions provided in the activity. Teachers should look for opportunities to develop students’ ability to justify methods. Students may need support from sentence stems to understand how to respond to their peers.</td></tr>
<tr><td>Differentiate Instruction</td><td colspan="2">Using the flow charts for deductive and inductive reasonings, consider creating a flow chart with examples of definitions, postulates, properties, theorems, a pattern, and a conjecture so visual learners can connect the process for future reference.</td></tr>
<tr><td colspan="3">Lesson Notes:</td></tr>
</table>

2.2.1 Warm-Up: What's the Pattern?

Component Overview: *Given the first four steps of a pattern with different colored circles, the students will anticipate the number of circles in the fifth step of the pattern. For question 4, students will describe the pattern observed.*

2.2.1 Warm-Up: What's the Pattern?

Four steps of a pattern are shown.

Step 1	Step 2
	
Step 3	**Step 4**
	

1. How many blue circles will be in step 5?
   33 *blue circles*
2. How many orange circles will be in step 5?
   16 *orange circles*
3. How many green circles will be in step 5?
   20 *green circles*
4. Describe the pattern using words or an algebraic expression.
   Answers will vary. Blue circles follow the pattern of $6n+3$*, where* n *is the step number in the pattern. Orange circles follow the pattern of* $(n-1)^2$*, where* n *is the step number in the pattern. Green circles follow the pattern of* $4n$*, where* n *is the step number in the pattern.*

If students do not write an algebraic expression for the pattern, avoid giving hints for doing this, as they will be writing this in the next component, 2.2.2 Exploration.

2.2.2 Exploration: Types of Reasoning

Component Overview: *Students will work together to complete an Exploration on different types of reasoning.*

2.2.2 Exploration: Types of Reasoning

To describe a pattern such as the pattern in the Warm-Up is to make a conjecture.

A ***conjecture*** is a mathematical statement that has not yet been proven.

Once a conjecture has been made, then reasoning can be used to prove the conjecture.

1. Work with your partner to write conjectures for the pattern in the Warm-Up. You should be able to use each conjecture to determine any step in the pattern.

    - number of blue circles
    - number of orange circles
    - number of green circles
    - total number of circles

Blue	Orange	Green	Total
$6n+3$ is the equation to find the number of blue circles, where n is the step number.	*$(n-1)^2$ is the equation to find the number of orange circles, where n is the step number.*	*$4n$ is the equation to find the number of green circles, where n is the step number.*	*n^2+8n+4 is the equation to find the total number of circles, where n is the step number.*

A conjecture is an important step in problem-solving. Conjectures arise when you notice a pattern and that pattern holds true for many cases. However, just because a pattern holds true for many cases does not mean that the pattern will hold true for ***all*** cases. Conjectures must be proven for the mathematical observation to be fully accepted. When a conjecture is rigorously proven, it becomes a theorem.

Students may not be familiar with the word "conjecture." If your students need further support consider beginning this lesson by introducing the concept of conjecture and release students to complete the activity up through the flow chart for deductive and inductive reasoning. Once you introduce those concepts, rerelease students to complete the remainder of the activity.

While students are working with their partner, monitor that students are moving forward with completing the table. The focus of the lesson is on developing a conjecture, not the algebra of writing the expression. Prepare questions prior to class that may assist them if they struggle with the table. Sample questions are:

- *Can you create a table for each color with the step number as the input and the number of circles as the output? What patterns do you notice for the blue and green circles? What type of numbers are the numbers of orange circles?*
- *How can you write a pattern using the step number as your input (variable)?*

2.2.2 Exploration: Types of Reasoning (continued)

The values for step 10 are shown. Use these values to test your conjecture. Revise your conjecture if necessary.

Blue	Orange	Green	Total
63	81	40	184

2. Ask a classmate for their conjectures. Record their conjectures and write your summary of their reason. *You are the only person who should write in the conjecture and summary columns of the table.* Have the person initial next to your summary stating that your summary was correct.
   Answers will vary.

Color	Conjecture	My Summary of Their Reason	Initials
Blue			
Orange			
Green			
Total			

The conjectures from your classmate may be different from your conjectures, but that does not mean that one is wrong. Both may be correct.

3. Compare your conjecture with the conjectures you and your partner collected from different classmates.

4. The values for step 18 are shown. Use these values to test your classmates' conjectures.

Blue	Orange	Green	Total
111	289	72	472

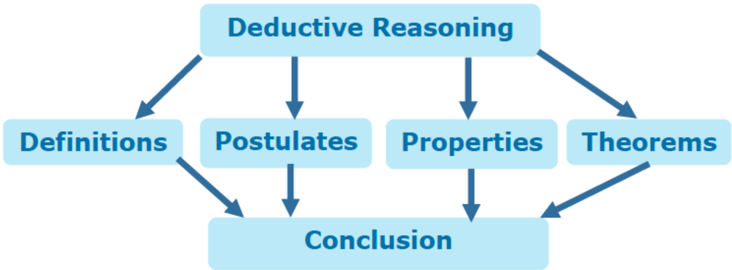
Answers will vary.

Students may need directions for question 2 in terms of logistics and a reminder of classroom expectations for their level of movement and conversation. If classroom behavior is an issue, consider assigning each student a specific partner within an appropriate proximity to avoid classroom disruption during this piece. The classmate for question 2 is not their original partner for the 2.2.2 Exploration.

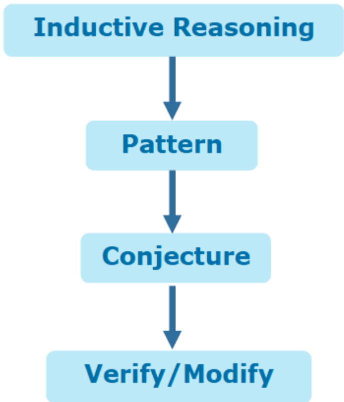
2.2.2 Exploration: Types of Reasoning (continued)

There are two types of reasoning that can be used to prove a conjecture.

Deductive reasoning uses facts and general principles to draw conclusions.



Inductive reasoning uses specific observations and patterns to draw conclusions.



Consider creating an anchor chart for this step that helps students have a visual reference when dealing with deductive and inductive reasoning.
For a note-taking reference guide, consider providing examples where things they learned in prior mathematics would be used as a form of deductive reasoning, such as the distributive property or triangle sum theorem.

5. Discuss with your partner which type of reasoning best describes the type that was used in the warm-up. Summarize your discussion.
   Answers will vary. The patterns are very much an inductive reasoning process, so the warm-up must be an example of inductive reasoning.

Much of the reasoning that will be used in geometry is deductive reasoning.

Deductive reasoning needs to be used to prove the following statement is true.

All numbers ending in 0 or 5 are divisible by 5.

To verify that ***all*** numbers that end in 0 or 5 are divisible by 5 would require more than the method of verifying and modifying conjectures. When trying to show that something works for all cases, deductive reasoning should be used.

6. The following statement is an example of inductive reasoning.
   All small dogs in the park today have blue collars.
   Therefore, all small dogs have blue collars.
   Discuss with your partner if it is possible that there existed a scenario that could have led to the conclusion. Write a short explanation.
   Answers will vary. The conclusion could have been drawn if all the small dogs in the park on a given day had on blue collars.

A ***counterexample*** is a statement that disproves a conjecture.

7. Write a counterexample that would disprove "all small dogs have a blue collar."
   Answers will vary. I know someone who has a small dog, and that dog does not have a blue collar.

Students often make the mistake of creating an inverse statement when trying to create a counterexample. Watch for this misconception as students work to create counterexamples throughout this and future lessons.

2.2.3 Bring It Together: Using Reasoning

Component Overview: *Students apply their learning from the prior activity to formalize the concept of using reasoning.*

2.2.3 Bring It Together: Using Reasoning

1. Determine the type of reasoning for each statement.

Statement	Type of Reasoning
The car's battery provides power to the engine, so if the battery is dead, the car won't start.	*Deductive*
I know Joe is a good cook because I've eaten at his house three times and each time the food has been amazing.	*Inductive*
The diary that is supposed to be from the 18th century might be real. It has passed a number of forensic and historical tests.	*Deductive*
The sum of two 2-digit numbers is always a 3-digit number.	*Inductive*

2. For each statement, write a counterexample that disproves the statement.

   a. If a number is multiplied by itself, then the product is even.
   $3 \cdot 3 = 9$ and 9 is not even.

   b. Adding the same number to both the numerator and the denominator of a fraction does not change the fraction's value.
   $\frac{3}{5} \neq \frac{3+2}{5+2} = \frac{5}{7}$

Use your informal observations from 2.2.2 Exploration to dictate how to summarize the learning of the Exploration. If students struggled with the concepts, then revisit the major points before beginning 2.2.3 Bring It Together. If only a few students struggled then consider using small group instruction for some while others attempt the questions independently or facilitate the component as a whole group.

Observe students as they complete the table for question 1. Their responses will provide data for what they understand and don't understand about deductive and inductive reasoning.
If you see a student with the answers inductive-deductive-inductive-deductive, it is a sign that they understand the reasoning types but have them backwards with identification.
If students have one answer incorrect, it indicates that something in the language of the statement caused confusion as to identifying the right type.
Any other combinations of incorrect answers are a sign that students may need additional remediation and support to understand the concept.

2.2.4 Wrap-Up: Cool Down

***Component Overview**: Students write a short summary of their learning in the lesson. Student responses can be used to identify misconceptions and plan for upcoming instruction.*

2.2.4 Wrap-Up: Cool Down

1. Write a statement that uses deductive reasoning.
   Answers will vary.

2. Write a counterexample to the statement.
   Answers will vary.

2.3 Logical Conclusions

Lesson Introduction

Benchmark	MA.912.LT.4.10 Judge the validity of arguments and give counterexamples to disprove statements.	Learning Targets	• I can complete an argument that leads to a logical conclusion. • I can determine if an argument leads to a logical conclusion.

	Building On	Working Toward
Benchmark Coherence	N/A	N/A

Essential Questions	• How can I know if a conclusion is logical? • In what ways can I determine if an argument leads to a logical conclusion?
Lesson Narrative	In this lesson, students will develop arguments that lead to a logical conclusion. Students will begin by collaborating to connect logical arguments with the game of UNO. Then, they will complete UNO proofs to draw a logical conclusion.

Component Summary	2.3.1 Warm-Up (~5 min.)	Bell Work using deductive and inductive reasoning *(SE p. 94)*	2.3.3 Activity (20 min.)	Create logical arguments using UNO cards *(SE p. 96)*
	2.3.2 Collaboration (20 min.)	Evaluate and create logical arguments using UNO cards *(SE p. 95)*	2.3.4 Wrap-Up (~5 min.)	Summary questions about 2.3.3 Activity *(SE p. 97)*

	Glossary	Materials	Additional Preparation
Resources	Key Terms: N/A Additional Terms: N/A	• UNO Proof Task Cards • (Optional) UNO Card Decks	• Each UNO proof in 2.3.3 Activity can vary in level of difficulty. Work through each proof ahead of time to determine if scaffolding may be necessary or if a specific order might be provided. Not all proofs are designed to be completed by all students. The number of proofs needed will depend on how you divide the students.

<table>
<tr><td rowspan="4">Teacher Tips</td><td>Common Misconceptions</td><td>Things to Consider</td></tr>
<tr><td>Students may not be familiar with the game UNO or may use alternate rules than the standard rules.
Students may not connect the reasons explicitly to each statement.
Students may miss a step in the proof process.</td><td>This lesson is designed to provide students with a connection between a common understanding of rules (i.e. through playing UNO) to completing a two-column proof through statements and accurate reasons.</td></tr>
<tr><td>Literacy and Language Routines</td><td>Mathematical Thinking and Reasoning (MTR) Standard</td></tr>
<tr><td>Clarify, Critique, Correct
2.3.2 Collaboration question 7 provides students an opportunity to take a partial logical argument and fill in the appropriate blanks to make the argument work. Leveraging a Clarify, Critique, Correct protocol can help students work together to successfully complete the logical two-column argument.</td><td>MA.K12.MTR.1.1: Participate in Effortful Learning
This lesson is centered around collaborative activities among student pairs or small groups. In 2.3.2 Collaboration, students work together to connect UNO steps to constructing a logical argument. In 2.3.3 Activity, students work together to apply that learning to completing UNO proofs. Utilize these activities to cultivate a community of growth mindset learners and to foster perseverance in students.</td></tr>
<tr><td>Differentiate Instruction</td><td colspan="2">This lesson provides UNO proofs for students to complete. Consider scaffolding some of the proofs to add more completed steps or remove previously completed steps to increase or decrease the cognitive demand of the task itself based on student need.</td></tr>
<tr><td colspan="3">Lesson Notes:</td></tr>
</table>

2.3.1 Warm-Up: Bell Work

***Component Overview:** Students complete a Bell Work activity over the prior lesson: deductive reasoning, inductive reasoning, and counterexample.*

2.3.1 Warm-Up: Bell Work

1. Students at Cypress Bay High School must have a 2.0 grade point average in order to participate in sports. Oliver has a 2.0 grade point average. Oliver can play lacrosse for the Cypress Bay Lighting.

   Determine whether the scenario uses deductive or inductive reasoning.
   Deductive reasoning.

2. Write a counterexample for the given statement.

   If $\angle CRK$ is an acute angle, then $m\angle CRK = 84°$.

   Answers will vary.
   Any statement of $m\angle CRK$ set equal to any value between $0°$ and $90°$ that is not $84°$ is correct.
   Example: $m\angle CRK = 52°$

Use student understanding to inform your instruction and to determine where during this and further lessons small group instruction could be utilized. Make a list of misconceptions and determine which should be addressed whole group and which should be addressed in a small group with those who are struggling to be proficient.

2.3.2 Collaboration: Logical Arguments Using UNO

Component Overview: *Students will collaborate to connect the concept of logical arguments to the game of UNO. For this activity, students will work together in partner pairs or in small groups.*

2.3.2 Collaboration: Logical Arguments Using UNO

In the game of UNO there are 112 cards. There are four sets of color cards: blue, green, red, and yellow, with digits 0 to 9. Each color has three types of specialty cards: Skip, Reverse, and Draw 2. There are two types of Wild cards: Wild and Wild Draw 4.

For each player's turn, the player matches the top card on the discard pile either by number, color, or word. If the player doesn't have anything that matches, they must pick a card from the draw pile. If the player draws a card that can be played, then the player plays that card.

1. Work with your partner to determine if the series of steps is possible in the game. Each statement represents a card played.
   Yes, this series of steps is possible.

Statement	Reason
1. Green 8	1. Given
2. Blue 8	2. Matches number
3. Blue 3	3. Matches color
4. Wild	4. Changes color to yellow
5. Yellow Skip	5. Matches color

2. Add three additional statements and provide the reason.

Statement	Reason
6. *Yellow 2* *Answers will vary.*	6. *Matches color* *Answers will vary.*
7. *Yellow 5* *Answers will vary.*	7. *Matches color* *Answers will vary.*
8. *Red 5* *Answers will vary.*	8. *Matches number* *Answers will vary.*

Prior to releasing students to work on their own with their partner or group, review the narrative language that is provided prior to question 1. This will help set the stage for the activity.

Be mindful that some students may not have played the game UNO before or may play by a different set of rules than the standard UNO rules. Consider surveying students to determine what they know and do not know about the game. Show a basic review of the game if necessary.

This collaboration is scaffolded to allow students to engage with it with their partner rather than being guided through by the teacher. The idea of proofs is often intimidating for students. This lesson is meant to develop students' understanding of a logical argument and how one step in a proof leads to the next (in most cases). Allow students to struggle and to revise their thinking throughout the collaboration. Observe and listen as students work with their partners while providing a safe environment for the students to be wrong and to ask questions. Additional scaffolding questions are provided, if needed, but this should be primarily student-directed.

Why do you think it is possible for there to be more than one correct answer for the next step? Why are "Yellow 2" and "Blue Skip" both correct answers? Would both correct answers have the same reason? Why or why not?

2.3.2 Collaboration: Logical Arguments Using UNO (continued)

3. Ask your partner to verify that your steps are valid.

4. The series of steps shown is not possible. Correct the error.
Answers will vary.

Statement	Reason
1. Red 4	1. Given
2. Red 9	2. Matches color
3. Yellow 9	3. Matches number
4. Green ~~4~~ 9	4. ~~Changes color to green~~ *Matches Number*
5. Red ~~4~~ 9	5. Matches number
6. Red Reverse	6. Matches color

5. Ensure that you and your partner found the same error and that the corrections are both correct.

Yolanda made the following corrections.

Statement	Reason
1. Red 4	1. Given
2. Red 9	2. Matches color
3. Yellow 9	3. Matches number
4. ~~Green 4~~ Blue 9	4. ~~Changes color to green~~ Matches number
5. ~~Red 4~~ Blue Skip	5. Matches ~~number~~ color
6. ~~Red Reverse~~ Blue 5	6. Matches color

If available, consider using physical UNO cards as part of this activity. This can help kinesthetic learners be able to engage in a hands-on experience while allowing visual learners to see the statement in action and connect the correct reason from the card game to the table.

For an additional challenge, have students create another series of steps that are not possible for their partner to find the error and make corrections.

2.3.2 Collaboration: Logical Arguments Using UNO (continued)

6. Explain why you or your partner's corrections may not match Yolanda's corrections.
   Answers will vary. There are many options for how this could be corrected. My partner changed the Green 4 to a Yellow 3 and had a completely different set of reasons as well.

7. Work with your partner to create a logical argument that leads to a Red Draw 2.
   Answers will vary.

Statement	Reason
1. Green 5	1. Given
2. Yellow 5	*2. Matches number*
3. Yellow $\boxed{4}$	*3. Matches color*
4. *Green 4*	*4. Matches number*
5. Green 9	*5. Matches color*
6. Green 8	*6. Matches color*
7. Green 3	*7. Matches color*
8. Red 3	*8. Matches number*
9. Red 8	*9. Matches color*
10. Red Draw 2	*10. Matches color*

8. With your partner, use the template provided by your teacher to fill in your reasons for 5, 8, and 10. Then switch your paper with another pair. Work with your partner to fill in the missing statements and reasons on the other pair's paper.
   Answers will vary.

9. Work as a group of four to make sure that the two completed templates correctly lead to the conclusion.

As you circulate consider asking students to share their answers to question 6. This question can vary based on how each pair chose to fix the original errors. Include in the discussion how it is possible for there to be multiple correct ways to fix the error and the importance of having an accurate reason that supports the statement. Use the opportunity to listen and build student confidence.

Consider using the digital template to provide scaffolds for students who may need support or to increase the challenge of this activity for students who are ready. This can be done by removing statements or changing the color provided, or only providing the given and the last step. Whichever you choose, work through it again to ensure that the answer will work with the number of statements provided in the template.

2.3.3 Activity: Taking Turns

Component Overview: *Students will work together to complete a series of proofs using UNO cards. For this activity, students should work in partner pairs or in small groups. The templates for the cards are located in the teacher guide.*

2.3.3 Activity: Taking Turns

Complete the UNO proofs. All UNO cards are available except for the Wild or Wild Draw 4 card. The only time a Wild card can be used is when it is already written in the proof.

Each proof must have at least three number changes and three color changes.
Answers will vary.

For this activity, it is suggested that students work with partners and that each pair starts a proof. Then, the proof is passed to the next pair, etc. The goal is for all proofs to be completed by the class, not that each student pair completes each proof. Use the activity to cultivate a community of growth mindset learners and to foster perseverance by reminding the class that the goal is to work together to complete as many proofs as possible. For classes where students may struggle consider lowering the number of proofs and changing from partners to groups of three or four. For Honors, utilize all of the proofs or create some on your own that have an increased difficulty.
Alternative Methods:

- *Post the templates up around the room and have students complete one, two, or three steps before moving to the next.*
- *Have each pair of students complete a proof in its entirety and then share with the class.*

2.3.4 Wrap-Up: Reflection

Component Overview: *In this activity, students will answer three questions to summarize the activity. These questions can be done independently or with their partner or group from the prior activity.*

2.3.4 Wrap-Up: Reflection

1. How many of the UNO proofs did the class correctly complete?
   Answers will vary.

2. Which steps in the UNO proof were the easiest to complete?
   Answers will vary.

3. Which steps in the UNO proof were hardest to complete?
   Answers will vary.

The information provided from the student here, coupled with what was observed during instruction, can provide information for how engaged each student was during the lesson. This provides an individual accountability of a student's engagement (MA.K12.MTR.1.1). Consider reviewing student responses to consider which students may need further support in effortful learning. Also consider writing an encouraging note to students about their engagement.

2.4 Hypothesis and Conclusion

Lesson Introduction

Benchmark	MA.912.LT.4.3 Identify and accurately interpret “if…then,” “if and only if,” “all” and “not” statements. Find the converse, inverse, and contrapositive of a statement.	Learning Targets	• I can identify the hypothesis of a statement. • I can identify the conclusion of a statement. • I can write the converse, inverse, and contrapositive of a conditional statement.

Benchmark Coherence	Building On	Working Toward
	N/A	N/A

Essential Questions	• What is the relationship between a hypothesis and a conclusion of a statement? • How can I write the converse, inverse, and contrapositive of a conditional statement?
Lesson Narrative	In this lesson, students are introduced to the concept of hypothesis and conclusion within the written form of a conditional statement. Students will engage in an Exploration of what happens when the hypothesis and the conclusion are switched. The new vocabulary “converse,” “inverse,” and “contrapositive” are introduced with a practice opportunity of writing statements of each.

Component Summary				
	2.4.1 Warm-Up (~5 min.)	Career highlights of a prototype fabricator *(SE p. 98)*	2.4.4 Guided Instruction (15 min.)	Writing converse, inverse, and contrapositive statements *(SE p. 100)*
	2.4.2 Guided Instruction (10 min.)	Writing conditional statements and identifying the hypothesis and conclusion *(SE p. 98)*	2.4.5 Guided Practice (10 min.)	Writing the converse, inverse, and contrapositive for conditional statements *(SE p. 101)*
	2.4.3 Exploration (5 min.)	Exploring what happens when the hypothesis and conclusion are switched *(SE p. 99)*	2.4.6 Wrap-Up (~5 min.)	Cool Down on conditional statements and their converse, inverse, and contrapositive *(SE p. 101)*

Resources	Glossary	Materials	Additional Preparation
	<u>Key Terms:</u> • Conditional statement • Converse statement • Inverse statement • Contrapositive statement <u>Additional Terms:</u> • Hypothesis • Conclusion	N/A	• With these concepts, students often need repetitive practice and examples to help them understand. When reviewing the lesson, consider developing a few additional examples of each that would be relatable to your specific students. • Warm-Up 2.4.1 contains a video to accompany the question prompts.

<table>
<tr><td rowspan="4">Teacher Tips</td><td>Common Misconceptions</td><td>Things to Consider</td></tr>
<tr><td>• Students may write the inverse statement as the converse or vice-versa.
• Students may forget to switch the hypothesis, p, and the conclusion, q, when writing a contrapositive statement.
• Students may struggle with the notation in the definitions using the hypothesis, p, and the conclusion, q, without a context.
• Students may assume that statement written as conditionals are always true.</td><td>• This lesson can be very engaging for students. Students often confuse the relationship of the original statement to its converse, inverse, and contrapositive.
• Consider asking students if statements and their converses, inverses, and contrapositives are true. Although this is not a requirement of the benchmark it does set up students' understanding of biconditionals.</td></tr>
<tr><td>Literacy and Language Routines</td><td>Mathematical Thinking and Reasoning (MTR) Standard</td></tr>
<tr><td>Co-Craft Questions and Problems
2.4.4 Guided Instruction introduces converse, inverse, and contrapositive. Consider leveraging this routine to help students better understand these concepts. Allow students to create their own converse, inverse, and contrapositive statements at each step of the way.</td><td>MA.K12.MTR.7.1: Real-World Contexts
In 2.4.2 Guided Instruction and 2.4.4 Guided Instruction, students will be engaged in learning about hypotheses and conclusions as well as converse, inverse, and contrapositive statements. Examples of these statements include real-world examples and quotes from famous people to provide for a more relatable entry for students into the concepts.</td></tr>
<tr><td>Differentiate Instruction</td><td colspan="2">This lesson is heavy on new vocabulary and introduces students to three concepts: converse, inverse, and contrapositive statements. These can be easy for students to get mixed up. Consider providing a Frayer Model template to allow students to have a reference for future use.</td></tr>
<tr><td colspan="3">Lesson Notes:</td></tr>
</table>

2.4.1 Warm-Up: Prototype Fabricator

Component Overview: *Play the Career Profile video on the prototype fabricator for the entire class. After students watch the video (3:48 minutes), prompt them to reflect on the profession by answering the two questions independently.*

2.4.1 Warm-Up: Prototype Fabricator

Jeff Tiedeken is a prototype fabricator. He creates metal prototypes through bending, cutting, welding, and assembling.

Jeff states that most people who come in for help do not have an idea of what they need, so his first job is to understand the problem.

1. How does understanding the problem help?
   Answers will vary. Understanding the problem helps him think of ways to solve the problem.

Jeff said that once he understands the problem, he then solves the problem with other people.

2. Why do you think discussing a problem with others can be helpful?
   Answers will vary. Discussing the problem with others allows him to see solutions from other perspectives.

Many students may not know what a prototype fabricator does. Students have many interests and talents yet they may not be aware of various careers. Notice that the follow-up questions purposively do not ask about how Jeff Tiedeken uses mathematics. Students should not be pressured to think that they cannot have a successful career without mathematics. Consider refraining from making this or other career profiles about mathematics. For students who dislike or struggle with mathematics, they may consider these careers as off-limits to them.

2.4.2 Guided Instruction: Hypothesis and Conclusion

Component Overview: *Students will engage in guided instruction on understanding a conditional statement and both a hypothesis and a conclusion. For this activity, consider utilizing a gradual release model through each of the questions.*

2.4.2 Guided Instruction: Hypothesis and Conclusion

Often information is given as an if-then statement. A few examples are shown.

- If you leave the lights on, then it will take longer for people to fall asleep.
- If a goldfish is not exposed to light, then it will lose its color.
- If a quadrilateral has 4 right angles, then it is a rectangle.

1. Write a new if-then statement using a different phrase following "if."
   Answers will vary.

   If *people are on their cell phones at night*, then it will take longer for people to fall asleep.

2. Write a new if-then statement using a different phrase following "then." *Answers will vary.*

   If a quadrilateral has 4 right angles, then *it is not a kite*.

A ***conditional statement*** is a statement in the format of "If P, then Q." The phrase P is the hypothesis. The phrase Q is the conclusion.

3. What is the hypothesis for "If a quadrilateral has 4 right angles, then it is a rectangle"?
   A quadrilateral has 4 right angles.

4. Write a conclusion for each hypothesis.
   Answers will vary.
   a. If yesterday was Sunday, then *today is Monday*.
   b. If you see lightning, then *there is a storm nearby*.

5. William Camden wrote, "The early bird catches the worm." Rewrite Camden's statement as a conditional statement.
   If the bird is early, then it catches the worm.

Discussion questions to consider:

- *Does an if-then statement have to be true?*
- *What is another example of an if-then statement?*
- *Does every if-then statement have to contain both components? Why or why not?*
- *Can the "then" statement exist without there being an "if" statement? Why or why not?*

Understanding the idea of "conditional statement" is an important concept of this lesson. Consider spending extra time reviewing this concept with students to ensure they have a solid understanding, including which part is the hypothesis and which part is the conclusion. Consider adding additional examples that are relatable to your specific students. Consider having students evaluate the truth of a given conditional statement, as all statements written as conditionals are not always true.

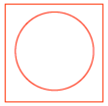
2.4.3 Exploration: Switching the Hypothesis and Conclusion

Component Overview: *Students will explore what happens when the hypothesis and the conclusion of a conditional statement are switched. For this activity, students should complete Questions 1 and 2 on their own and then work with their partner on Question 3.*

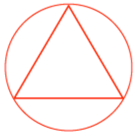
2.4.3 Exploration: Switching the Hypothesis and Conclusion

1. Draw a square or a triangle.

2. If you drew a square, then draw a circle inside the square.



   If you drew a triangle, then draw a circle outside the triangle.



3. Discuss with your partner what happens when the hypothesis and conclusion of each conditional are switched.

   - If you drew a circle inside the square, then draw a square.
   - If you drew a circle outside the triangle, then draw a triangle.

   Summarize your discussion.
   Answers will vary. If you switch the hypothesis, and conclusion, these statements do not make much sense. If they are reworded, then you may be able to still get the same shape. It is not possible to draw a circle inside a square that does not exist yet. It is not possible to draw a circle outside a triangle that does not exist yet.

What students discover during this activity is going to serve as the bridge between what they just learned about conditional statements and what they are about to learn in the next activity with converse, inverse, and contrapositive statements. Do not give away those terms yet in this activity.
Allow a few minutes for students to share out their answers to question 3. This will provide formative data on whether or not students got the intended outcome of this activity. Use this discussion to clear up any misconceptions that may arise.

For students who grasp this quickly and finish early, provide them with a challenge of writing a statement that is true both in the original conditional and when switched.

2.4.4 Guided Instruction: Converse, Inverse, and Contrapositive

Component Overview: *Students are introduced to the vocabulary of converse, inverse, and contrapositive statements. For this activity, engage students in the gradual release model as you guide them through the content of this activity.*

2.4.4 Guided Instruction: Converse, Inverse, and Contrapositive

The ***converse statement*** is a statement in which the given conditional, "If P, then Q," changes to "If Q, then P."

1. Write the converse of "If it is raining, then the grass is wet."
   If the grass is wet, then it is raining.

2. Write a counterexample that shows the converse is not true.
   Answers will vary. A sprinkler may have just been turned off after watering the grass.

The ***inverse statement*** is a statement in which the given conditional, "If P, then Q," changes to "If not P, then not Q."

3. Complete the inverse for "If it is raining, then the grass is wet."
   If it is *not* raining, then the grass is *not* wet.

4. Write a counterexample that shows the inverse is not true.
   Answers will vary. If someone is washing a car on the grass, it may not be raining, however the grass will be wet.

The ***contrapositive statement*** is a statement in which the given conditional, "If P, then Q," changes to "If not Q, then not P."

5. Write the contrapositive for "If it is raining, then the grass is wet."
   If the grass is not wet, then it is not raining.

This section contains several vocabulary terms. Consider the use of a foldable, Frayer Model, anchor chart, or other type of interactive notebook for this section. Whichever method is chosen, it will be important that examples be provided so that students can refer back to them in the future.

While the lesson has specific examples to accompany each statement type, students may need additional examples to connect the content. Prior to releasing students to complete questions 2, 4, and 5, consider providing additional examples that your students can connect to. This can include statements about your town, your school, a famous athlete, a catchy ad slogan, or other relevant examples.

2.4.5 Guided Practice: Converse, Inverse, and Contrapositive

Component Overview: *Students will write the converse, inverse, and contrapositive statements for given conditionals. For this activity, allow students to work independently but check their work with a partner.*

2.4.5 Guided Practice: Converse, Inverse, and Contrapositive

Write the converse, inverse, and contrapositive for each conditional.

1. "If you're not making mistakes, then you're not doing anything." (Quote by John Wooden, an American basketball player and coach)

Converse	*If you're not doing anything, then you're not making mistakes.*
Inverse	*If you are making mistakes, then you are doing something.*
Contrapositive	*If you are doing something, then you are making mistakes.*

2. "If you think you're always in control, then you're not going fast enough." (Quote by Mario Andretti)

Converse	*If you're not going fast enough, then you think you're always in control.*
Inverse	*If you don't think you're always in control, then you are going fast enough.*
Contrapositive	*If you're going fast enough, then you don't think you're always in control.*

Launch this activity with the appropriate level of scaffolding based on student understanding from the prior activity. For example, if most students are struggling with the concepts, consider modeling an example of one of the three in the practice and then releasing. If students generally did well, release them to try them on their own.
Consider implementing a Think-Pair-Share protocol for this activity. Allow students to try the problem on their own and then check their work with a partner.

Use the time for this activity to circulate the room to see how students are completing this table. Try to see as many in question 1 as possible, as any errors students make in that question, they are likely to make in question 2.

2.4.6 Wrap-Up: Cool Down

Component Overview: *Students will complete a Cool Down that assesses their understanding of conditional statement, hypothesis and conclusion, and converse, inverse, and contrapositive statements. This activity should be done independently to provide specific student-level data for additional support.*

2.4.6 Wrap-Up: Cool Down

1. Rewrite the following quote by Michelle Obama as a conditional statement.

   "When someone is cruel or acts like a bully, you don't stoop to their level."
   If someone is cruel or acts like a bully, then you don't stoop to their level.

2. Complete the table for the conditional statement shown.

   If you have ridden the roller coaster SheiKra, then you have been to Busch Gardens.

Hypothesis	*You have ridden the roller coaster SheiKra.*
Conclusion	*You have been to Bush Gardens.*

3. Write the converse, inverse, and contrapositive for the conditional statement shown.

   If a fruit is yellow, then it is a banana.

Converse	*If a fruit is a banana, then it is yellow.*
Inverse	*If a fruit is not yellow, then it is not a banana.*
Contrapositive	*If a fruit is not a banana, then it is not yellow.*

This Cool Down leverages each of the three questions to assess a different component of this lesson. Based on student performance, you can determine which areas, if any, need additional support and with which students.

- *Question 1: Conditional Statements*
- *Question 2: Hypothesis and Conclusion*
- *Question 3: Converse, Inverse, and Contrapositive statements*

2.5 Conditionals and Biconditionals

<table>
<tr><th colspan="6">Lesson Introduction</th></tr>
<tr><td>Benchmark</td><td colspan="2">MA.912.LT.4.3 Identify and accurately interpret “if…then,” “if and only if,” “all” and “not” statements. Find the converse, inverse, and contrapositive of a statement.</td><td>Learning Targets</td><td colspan="2">I can write the converse, inverse, and contrapositive of a mathematical statement.
I can determine if a statement is biconditional.
I can write a definition as a biconditional.</td></tr>
<tr><td rowspan="2">Benchmark Coherence</td><td colspan="2">Building On</td><td colspan="3">Working Toward</td></tr>
<tr><td colspan="2">N/A</td><td colspan="3">N/A</td></tr>
<tr><td>Essential Questions</td><td colspan="5">How can I determine if a statement is biconditional?
How can I determine if a statement written in “if…then” form is true?</td></tr>
<tr><td>Lesson Narrative</td><td colspan="5">In this lesson, students continue their work with conditional, converse, inverse, and contrapositive statements. Students will engage in an activity where they determine the truth of the converse, inverse, and contrapositive of a given true conditional statement. They will then take what they observed in the activity to develop an understanding of a biconditional statement and its relationship to definitions. Biconditional statements will be written in “if and only if” form.</td></tr>
<tr><td rowspan="3">Component Summary</td><td>2.5.1 Warm-Up
(~5 min.)</td><td>Calculating music streaming payouts (SE p. 102)</td><td>2.5.3 Bring It Together
(10 min.)</td><td colspan="2">Connecting true conditionals and converses to biconditional statements and definitions (SE p. 104)</td></tr>
<tr><td>2.5.2 Activity
(20 min.)</td><td>Determining the truth value of conditionals, converses, inverses, and contrapositives (SE p. 103)</td><td>2.5.4 Your Turn!
(10 min.)</td><td colspan="2">Writing and determining the truth value of statements to determine if a biconditional exists (SE p. 105)</td></tr>
<tr><td>2.5.5 Wrap-Up
(~5 min.)</td><td colspan="4">Self-assessment on student confidence with each concept from this lesson (SE p. 106)</td></tr>
<tr><td rowspan="2">Resources</td><td colspan="2">Glossary</td><td>Materials</td><td colspan="2">Additional Preparation</td></tr>
<tr><td colspan="2"><u>Key Terms:</u>
Biconditional
Definition
<u>Additional Terms:</u>
Converse
Inverse
Contrapositive
Truth Value</td><td>N/A</td><td colspan="2">Work each of the problems in 2.5.2 Activity prior to the lesson. Determine if you may need to provide additional background knowledge to support the claims made in the conditional statements, as some students may not be familiar with these facts.</td></tr>
</table>

<table>
<tr><td rowspan="4">Teacher Tips</td><td>Common Misconceptions</td><td>Things to Consider</td></tr>
<tr><td>Students may write the inverse statement as the converse or vice-versa.
Students may forget to switch the hypothesis, p, and the conclusion, q, when writing a contrapositive statement.
Students may mistake the definition of biconditional and forget to confirm the converse is also true.
Students may assume that statements written as conditionals are always true.</td><td>Pacing will be a key factor in this lesson, especially with the 2.5.2 Activity. This activity has 10 conditional statements, each with a converse, inverse, and contrapositive truth value. Work through each problem and time yourself to determine whether you want all students to do all 10 or if you will need to divide them up between groups.</td></tr>
<tr><td>Literacy and Language Routines</td><td>Mathematic Thinking and Reasoning (MTR) Standard</td></tr>
<tr><td>Poll the Class
2.5.2 Activity provides students with the opportunity to determine if the converse, inverse, and contrapositive of a conditional statement are true or false. Upon completion of each statement, consider polling the class to determine whether they believe it is true or false.</td><td>MA.K12.MTR.6.1: Reasonableness
2.5.2 Activity has students determine whether the converse, inverse, and contrapositive of a conditional statement are true or false. Students are given various true conditional statements grounded in geometry. Students then evaluate the resulting statement to determine its truth. The activity not only serves to build understanding of converse, inverse, and contrapositive of a conditional statement but also builds a student’s understanding of geometric concepts that then leads to understanding a biconditional.</td></tr>
<tr><td>Differentiate Instruction</td><td colspan="2">This lesson is heavy on new vocabulary and continues using the vocabulary: converse, inverse, and contrapositive statements. These can be easy for students to get mixed up. Consider providing a Frayer Model template to allow students to have a reference for future use.</td></tr>
<tr><td colspan="3">Lesson Notes:</td></tr>
</table>

2.5.1 Warm-Up: Streaming Payouts

Component Overview: *Students will analyze an image that shows streaming payouts per platform. They will use this information to answer questions on how much a payout will be for a number of streams and how many streams are needed for a particular payout.*

2.5.1 Warm-Up: Streaming Payouts

The payouts to a musician for each time a song is streamed on multiple streaming platforms in 2019 is shown.

Platform	Payout Per Stream
Napster	$0.01900
Tidal	$0.01250
Apple Music	$0.00735
Google Play	$0.00676
Deezer	$0.00640
Spotify	$0.00437
Amazon	$0.00402
Pandora	$0.00133
YouTube	$0.00069

If time permits, allow students to share strategies they used to complete the table. Highlight the strategy of using a power of ten to quickly determine the payouts.

1. Determine the payout for a musician whose song was streamed 100,000 times on each platform.

Platform	Payout for 100,000 Streams (in dollars)
Napster	$1,900
Tidal	$1,250
Apple Music	$735
Google Play	$676
Deezer	$640
Spotify	$437
Amazon	$402
Pandora	$133
YouTube	$69

2. How many streams on Spotify did it take to for a musician to make $1 in 2019?
   229 *streams*

2.5.2 Activity: True or False

Component Overview: *Students will complete an activity where they are given a conditional statement and must determine whether or not its converse, inverse, and contrapositive statements are true or false. For this activity, students can work independently or with a partner or small group.*

2.5.2 Activity: True or False

For each conditional statement given, consider whether its converse, inverse, and contrapositive statements are true or false. Record the results in the table for each conditional.

1. **Conditional**: If all of the angles of a triangle are less than $90°$, then the triangle is an acute triangle.

Relationship to Conditional	True or False
Converse	*True*
Inverse	*True*
Contrapositive	*True*

2. **Conditional**: If the sum of two angles is $90°$, then the angles are complementary.

Relationship to Conditional	True or False
Converse	*True*
Inverse	*True*
Contrapositive	*True*

3. **Conditional**: If a polygon has six sides, then it is a hexagon.

Relationship to Conditional	True or False
Converse	*True*
Inverse	*True*
Contrapositive	*True*

Timing will be a key consideration when doing this activity.
This activity has 9 conditional statements, each with a converse, inverse, and contrapositive truth value. Work through each problem and time yourself to determine whether you want all students to do all 9 or divide them between groups.
Because each conditional is a definition the converse, inverse, and contrapositive statements are true.
Students may second-guess themselves. The activity was purposely created to set up an understanding of biconditionals. If in Lesson 4 it was established that not all converse, inverse, and contrapositive statements of a conditional are true then students may understand the point more. If this was not done then consider using a few examples from Lesson 4 before releasing students to do the activity.

If time permits, consider asking students to include the actual statement with each type. This can be based on data from a previous lesson. For example, if you noticed most students struggled with inverse, have them write the inverse statement as part of this activity for each question.

2.5.2 Activity: Conditionals and Truth Values (continued)

4. **Conditional**: If a line or ray divides an angle into two equal angles, then the line or ray is the bisector of the angle.

Relationship to Conditional	True or False
Converse	True
Inverse	True
Contrapositive	True

5. **Conditional**: If lines that lie on the same plane do not intersect, then the lines are parallel.

Relationship to Conditional	True or False
Converse	True
Inverse	True
Contrapositive	True

6. **Conditional**: If the opposite sides of a quadrilateral are parallel, then the quadrilateral is a parallelogram.

Relationship to Conditional	True or False
Converse	True
Inverse	True
Contrapositive	True

7. **Conditional**: If $\overrightarrow{XY}$ is in the interior of $\angle WXZ$, then $m\angle WXY + m\angle YXZ = m\angle WXZ$.

Relationship to Conditional	True or False
Converse	True
Inverse	True
Contrapositive	True

Activity Facilitation Tip:

- *The answers to these truth-value questions are all "true." As you are circulating the room, you can determine if a student has a misconception by seeing the word "false."*
- *When addressing potential misconceptions with students, first ask them to tell you how the statement in question should be written. If they applied the wrong definition to it, then the misconception is with understanding converse, inverse, or contrapositive. If students have the statement correct, their misconception may be with the mathematical concept that is being described in the conditional statement.*

For question 7, students may not be familiar with the vocabulary "interior of an angle." This may need to be clarified prior to this question as the area between the two rays that define the angle.

2.5.2 Activity: Conditionals and Truth Values (continued)

8. **Conditional:** If two planes intersect, then their intersection is exactly one line.

Relationship to Conditional	True or False
Converse	*True*
Inverse	*True*
Contrapositive	*True*

9. **Conditional:** If two lines have the same slope, then they are parallel.

Relationship to Conditional	True or False
Converse	*True*
Inverse	*True*
Contrapositive	*True*

For question 8, students may not be familiar with intersecting planes. A quick discussion and demonstration of parallel and intersecting planes may be needed prior to this question.

For question 9, students will have to recall their knowledge of the equations of parallel lines from Algebra 1. Students may have forgotten what is true about the slopes of parallel lines. Consider a brief reminder/refresher on what is true about the slopes of parallel lines if needed.

2.5.3 Bring It Together: Biconditionals

Component Overview: *Students will take what they observed in the prior activity and bring it together to learn about the formal definition of biconditional statements. This activity can be done with a partner or small group but will need whole-group summary for a wrap-up. If needed, you can also deliver this through a Socratic seminar using the questions in the activity.*

2.5.3 Bring It Together: Biconditionals

1. How many of the conditionals from the previous Activity were considered true for all of the statements — converse, inverse, and contrapositive?
   All of the conditionals were considered true for all of the statements.

2. For how many conditionals was the converse of the conditional considered true?
   All of the conditionals were considered true for all of the converse statements.

A conditional statement is considered ***biconditional*** when both the statement and its converse are true. In other words, if both directions of the statement are true – "If P, then Q" and "If Q, then P" – the statement is biconditional.

Biconditional statements can be written in the format of "**if and only if**."

A **definition** is an exact statement, written formally, of the meaning or description of a word. Even though definitions in mathematics are often written as conditional statements, all definitions are actually biconditional statements.

3. Write each conditional using "if and only if."
   a. If the sum of two angles is $90°$, then the angles are complementary.
      The sum of two angles is 90° if and only if the angles are complementary.
   b. If the opposite sides of a quadrilateral are parallel, then the quadrilateral is a parallelogram.
      The opposite sides of a quadrilateral are parallel if and only if the quadrilateral is a parallelogram.

Consider a foldable, Frayer Model, or type of interactive notebook for this summary. Examples, non-examples, and verbal descriptions will provide multiple pathways for students to engage with the content in meaningful ways that encourage conceptual understanding over rote memorization.

2.5.4 Your Turn!

Component Overview: *Students will work independently to identify the converse, inverse, and contrapositive statements for a given conditional statement. They will then explain why the conditional statement is biconditional and write the conditional using "if and only if."*

2.5.4 Your Turn!

1. Complete the table for the conditional statement.

If all four angles at the point of intersection of two lines are right angles, then the lines are perpendicular.

Relationship to Conditional	Statement	True or False
Converse	*If two lines are perpendicular, then all four angles at the point of intersection are right angles.*	*True*
Inverse	*If all four angles at the point of intersection of two lines are not right angles, then the lines are not perpendicular.*	*True*
Contrapositive	*If two lines are not perpendicular, then all four angles at the point of intersection of two lines are not right angles.*	*True*

2. Explain why the conditional statement is biconditional.
   The conditional and the converse statement are both true, so the conditional qualifies as a biconditional.

3. Write the conditional using "if and only if."
   All four angles at the point of intersection of two lines are right angles if and only if the lines are perpendicular.

Use the data from the prior two activities to determine which students may need additional support. Some areas of support may be:

- *Writing converse statements*
- *Writing inverse statements*
- *Writing contrapositive statements*
- *Determining truth values for statements*
- *Understanding the conditions necessary to classify a statement as biconditional*

Do you think that a conditional statement can be biconditional if the converse is true but the contrapositive is not? Why or why not?

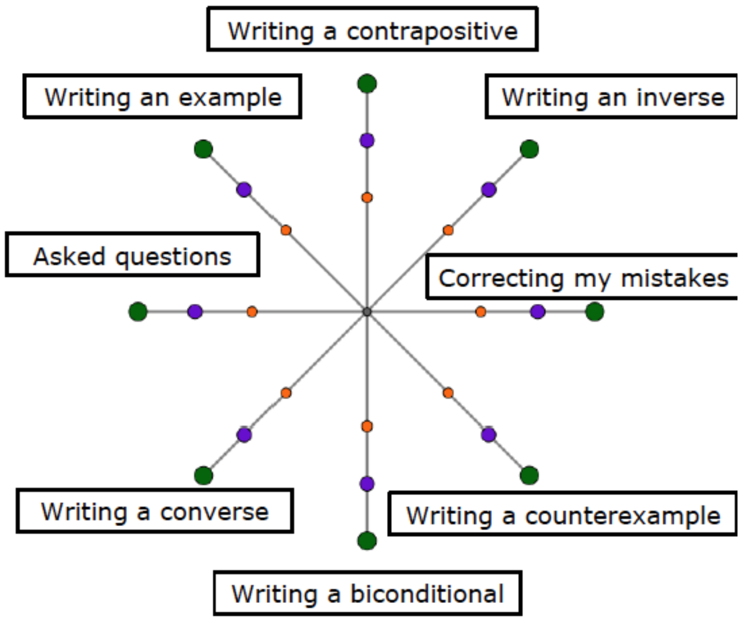
2.5.5 Wrap-Up: Reflection

Component Overview: *Students will complete a self-reflection on their level of confidence with eight components of this lesson. This activity should be done independently to get accurate student-level data on their perception of their own confidence related to this content.*

2.5.5 Wrap-Up: Reflection

1. Place an **X** on the circle that shows what you learned or completed during today's lesson.
   Answers will vary.

	I am confident I know how to do this!	OR	I always did this!
	I am fairly sure I know how to do this.	OR	I tried this.
	I need support on this.	OR	I had trouble with this.



The information provided from the student here, coupled with what was observed during instruction, can provide information for differentiating instruction in later lessons.

2.6 “All” and “Not” Statements

Lesson Introduction

Benchmark	MA.912.LT.4.3 Identify and accurately interpret “if…then,” “if and only if,” “all” and “not” statements. Find the converse, inverse, and contrapositive of a statement.	Learning Targets	• I can write and interpret an “all” statement. • I can write and interpret a negation.

	Building On	Working Toward
Benchmark Coherence	N/A	N/A

Essential Questions	• How can I know if an “all” statement is true or not? • What is the relationship between “all” and “not” statements?
Lesson Narrative	In this lesson, students will learn about “all” and “not” statements. Students examine the truth value of statements using the word “all” and edit them to use the word “some” if the truth value was false. They will explore how Venn diagrams can be used to represent mathematical or logical sets pictorially when comparing all and some. Students will also be introduced to the concept of the negation of a statement.

Component Summary	2.6.1 Warm-Up (~5 min.)	Bell Work on converse, inverse, contrapositive, and biconditional *(SE p. 107)*	2.6.3 Guided Practice (10 min.)	Determining truth values of “all” statements and rewriting statements as conditionals and negations *(SE p. 110)*
	2.6.2 Guided Instruction (30 min.)	Examining “all” and “not” statements *(SE p. 107)*	2.6.4 Wrap-Up (~5 min.)	Self-reflection on lesson concepts *(SE p. 111)*

	Glossary	Materials	Additional Preparation
Resources	<u>Key Terms:</u> • Venn diagram • Negation <u>Additional Glossary:</u> • Converse • Inverse • Contrapositive	• N/A	• For 2.6.2 Guided Instruction, consider displaying overlapping Venn diagrams and concentric Venn diagrams. Then discuss which type of Venn diagram would be used to represent the statements from question 4 for additional examples.

<table>
<tr><td rowspan="4">Teacher Tips</td><td>Common Misconceptions</td><td>Things to Consider</td></tr>
<tr><td>Students may write the inverse statement as the converse or vice-versa.
Students may forget to switch the hypothesis, p, and the conclusion, q, when writing a contrapositive statement.
Students may mistake a negation for another statement type, such as converse.
Students may assume that statements written in conditional form are always true.</td><td>This lesson requires the students to have a good understanding of the concepts in the prior three lessons. Consider using key moments in this lesson to provide additional support for students who need a review of that material.</td></tr>
<tr><td>Literacy and Language Routines</td><td>Mathematical Thinking and Reasoning (MTR) Standard</td></tr>
<tr><td>Stronger and Clearer Each Time
2.6.2 Guided Instruction introduces students to “all” and “not” statements. Provide students with your thinking as you help them move through the activity. This will allow students to model their thinking after yours as they are learning the new concepts.</td><td>MA.K12.MTR.5.1: Patterns and Structure
In 2.6.2 Guided Instruction, students are introduced to “all” and “not” statements. They will connect the structure of conditional statements and truth values to better understand the formal definition of an “all” statement and a negation.</td></tr>
<tr><td>Differentiate Instruction</td><td colspan="2">Throughout the lesson, if students are in need of differentiated support, consider modeling the first row of each table prior to releasing students to move forward with the remainder of the process as it unfolds.</td></tr>
<tr><td colspan="3">Lesson Notes:</td></tr>
</table>

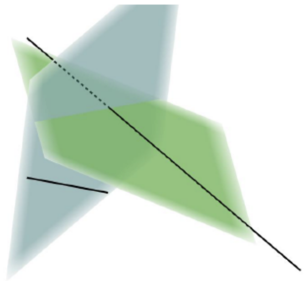
2.6.1 Warm-Up: Bell Work

Component Overview: *Students will complete a Bell Work covering the previous lesson. Students will write the converse, inverse, and contrapositive statements for a given conditional and determine their truth values. Students will also determine if the statement is a biconditional, writing the biconditional if it is or justifying why not if it is not.*

2.6.1 Warm-Up: Bell Work

1. Complete the table for the conditional statement.

If two lines are skew, then the lines do not lie on the same plane.



Relationship to Conditional	Statement	True or False
Converse	*If two lines do not lie on the same plane, then the lines are skew.*	*False*
Inverse	*If two lines are not skew, then the lines lie on the same plane.*	*False*
Contrapositive	*If two lines lie on the same plane, then the lines are not skew.*	*True*

2. If the conditional statement is biconditional, then write it as a biconditional. If the conditional statement is not biconditional, then explain why not.
The statement is not biconditional because the converse is false. (Skew lines do not intersect and are not parallel)

Skew lines are defined in Unit 1 Lesson 1 (G.1.1) if students need to reference the definition.

Use the data collected from this Bell Work to determine remediation needs and focused support during this lesson.

2.6.2 Guided Instruction: “All” and “Not” Statements

Component Overview: *Students will receive formal instruction on “all” statements and negation.*

2.6.2 Guided Instruction: “All” and “Not” Statements

Mathematical statements may not be conditional or biconditional. Mathematical statements could be written using “all.” The statement must be universally true to use the word “all.”

1. Discuss with your partner if the statement, “All quadrilaterals have four sides” is universally true. Summarize your discussion.
   Answers will vary. Yes, the definition of quadrilateral is that it has four sides.

2. Rewrite “All quadrilaterals have four sides” as an if-then statement.
   If a shape is a quadrilateral, then it has 4 sides.

3. Discuss with your partner why the statement “Some quadrilaterals have four sides” is not true. Summarize your discussion.
   Answers will vary. “Some” means there may be at least one quadrilateral that does not have only four sides. That is not true. There are no quadrilaterals that have a different number of sides.

4. Determine if each statement is true.

Statement	True or False
All dogs are Labradors.	*False*
All polygons have congruent sides.	*False*
All cats are mammals.	*True*

5. Rewrite each false statement using “some.”
   Some dogs are Labradors.
   Some polygons have congruent sides.

After questions 1 and 3, create opportunities for partner pairs to share out their answers along with rationale for everyone in the class to hear. Consider using an agree/disagree protocol in order to allow students to have a more productive dialogue with rationale and evidence.

How does changing a statement by adding the word “some” potentially change its truth value?

6. Write each statement as an if-then statement.

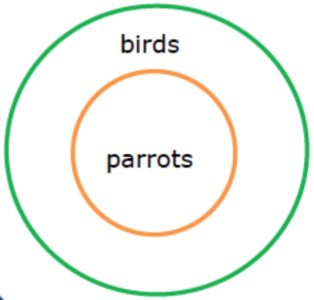
Statement	If-Then
All dogs are Labradors.	*If there are dogs, then they are Labradors.*
All polygons have congruent sides.	*If a figure is a polygon, then it has congruent sides.*
All cats are mammals.	*If an animal is a cat, then it is a mammal.*

It may be helpful to understand “all” statements using Venn diagrams.

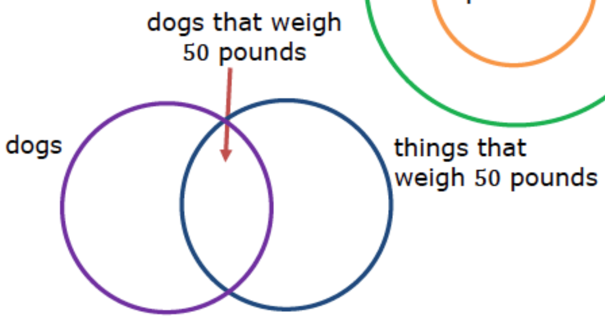
A **Venn diagram** is a diagram representing mathematical or logical sets pictorially as circles. Common elements of the sets are within the overlap of the circles.

7. The Venn diagram and a statement are shown.

All parrots are birds.



Some dogs weigh 50 pounds.



Consider using an anchor chart to help provide a visual example of the Venn diagram concept as it appears in this lesson. This way, students are able to refer back to it for future lessons when needed.

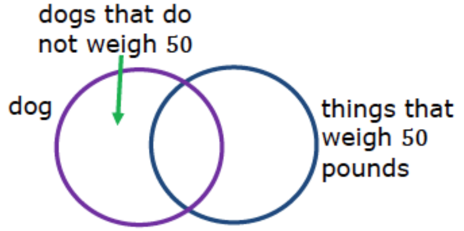
Students have learned about Venn diagrams in earlier grades and other content areas, but it may be beneficial to review the basic structure and function of a Venn diagram as part of this activity using an example that your specific students can easily relate to.

For more practice, represent the statements from question 4 using Venn diagrams.

2.6.2 Guided Instruction: "All" and "Not" Statements (continued)

Discuss with your partner how the Venn diagrams help to show the difference between "all" and "some." Summarize your discussion.
Answers will vary. If one circle (set) is drawn entirely inside of another, then "all" elements inside that circle (set) also belong to the larger circle (set). If two circles (sets) overlap, then "some" of the elements may belong in the overlapped part of both circles (sets) and "some" of the elements may belong in only one circle (set) but not the other, which is the nonoverlapped parts.

8. The Venn diagram below shows where dogs that do not weigh 50 pounds would be placed.



Write a statement that matches the Venn diagram.
Some dogs do not weigh 50 pounds.

The **negation** of a statement is the opposite of the statement. If the statement is true, then the negation must be false.

9. Use the definition of negation to write the negation of each statement.
   a. Today is Wednesday. *Today is not Wednesday.*

   b. Seo-jun is not wearing blue. *Seo-jun is wearing blue.*

The term "negation" may not be new for students, as it could have been used in other courses. It will be important for students to understand what is intended by the definition of negation in context of this concept.

Students may also confuse the negation of the statement as being synonymous with an inverse or converse statement. Consider front-loading this information when reviewing the terminology.

10. Rolando and Rhianna both wrote the negation of the statement, “$\sqrt{5}$ is an irrational number.”

Rolando	Rhianna
$\sqrt{5}$ is not an irrational number.	$\sqrt{5}$ is a rational number.

a. Discuss with your partner both negations. Write your observations and questions for the class.
Answers will vary. We discussed that each statement is the same. If a number is not an irrational number, then it is rational. So, to say a number is not irrational, is the same as saying that it is rational.

b. Based on the class discussion, which student has written a correct negation?
Both answers are mathematically equivalent.

Students learned about rational vs irrational numbers in a prior course. Depending on their middle school math progression, their exposure to the concept may be limited. Consider reviewing the concept between the two as part of reviewing question 10. Note that for Rhianna to be correct, there is an assumption that she is operating within the real-number system.

2.6.3 Guided Practice: "All" Statements and Negation

Component Overview: *Students will engage in a structured practice opportunity with "all" statements and negation. Consider the use of a gradual release model when facilitating this activity.*

2.6.3 Guided Practice: "All" Statements and Negation

1. Determine if each statement is true.

Statement	True or False
All congruent angles have the same measure.	*True*
All odd numbers are prime.	*False*
All points that lie in a plane are collinear.	*False*

2. Rewrite each false statement using "some."
   Some odd numbers are prime.
   Some points that lie in a plane are collinear.

3. Write each statement as an if-then statement.

Statement	If-Then
All congruent angles have the same measure.	*If angles are congruent, then they have the same measure.*
All odd numbers are prime.	*If a number is odd, then the number is prime.*
All points that lie on a plane are collinear.	*If points lie on the same plane, then the points are collinear.*

4. Write the negation of each statement.

Statement	Negation
ΔDTM is a right triangle.	*ΔDTM is not a right triangle.*
$m\angle TYW + m\angle WYK = 136°$	$m\angle TYW + m\angle WYK \neq 136°$
$\angle RWF$ and $\angle HYV$ are complementary angles.	*$\angle RWF$ and $\angle HYV$ are not complementary angles.*

With this activity, it is important to note that questions 1-3 are a continuation of the same scenario and are not three unique problems for students. If students need additional scaffolding with this concept, consider modeling one of the statements in question 1.

Question 4 is a unique question that is not part of the learning series of questions 1-3. Consider releasing students to work with a partner or small group to complete question 4.

To add physical movement, consider having students or student pairs find an additional student or pair to check their answers with.

2.6.4 Wrap-Up: Reflection

Component Overview: *Students will complete a self-reflection where they rate their level of understanding of concepts from this lesson. This activity should be done independently to provide an insight into each student's level of confidence with the content of this lesson.*

2.6.4 Wrap-Up: Reflection

1. Circle the icon that best describes your understanding of each concept below.
   Answers will vary.

   I need help. I got it. I could help someone.

I can write the converse of a mathematical statement.	
I can write the inverse of a mathematical statement.	
I can write the contrapositive of a mathematical statement.	
I can determine if a statement is biconditional.	
I can write and interpret an "all" statement.	
I can write and interpret a negation.	

The information provided from the student here, coupled with what was observed during instruction, can provide information for differentiating instruction in later lessons.

2.7 Proofs – Part 1

<table>
<tr><th colspan="5">Lesson Introduction</th></tr>
<tr><td>Benchmarks</td><td colspan="2">MA.912.GR.1.1 Prove relationships and theorems about lines and angles. Solve mathematical and real-world problems involving postulates, relationships, and theorems of lines and angles.

MA.912.LT.4.10 Judge the validity of arguments and give counterexamples to disprove statements.</td><td>Learning Targets</td><td>I can relate deductive reasoning to completing proofs based on an assumption or given.
I can recognize paragraph, flowchart, and two column proofs.</td></tr>
<tr><td rowspan="2">Benchmark Coherence</td><td colspan="2">Building On</td><td colspan="2">Working Toward</td></tr>
<tr><td colspan="2">MA.8.GR.1.4 Solve mathematical problems involving the relationships between supplementary, complementary, vertical or adjacent angles.</td><td colspan="2">N/A</td></tr>
<tr><td>Essential Questions</td><td colspan="4">How can I apply what I've learned about deductive reasoning to complete proofs?
What are the differences between the three different proof types?</td></tr>
<tr><td>Lesson Narrative</td><td colspan="4">This lesson is the first of a two-part lesson series on proofs. In this lesson, students connect their learning from the prior lessons to begin formally writing proofs. They will be introduced to three formal types of proofs: paragraph proofs, two-column proofs, and flowchart proofs. Students will compare the three types of proofs presented for the same given and conclusion.</td></tr>
<tr><td rowspan="3">Component Summary</td><td>2.7.1 Warm-Up (~5 min.)</td><td>Observing symmetry in Art Deco (SE p. 112)</td><td>2.7.3 Guided Instruction (10 min.)</td><td>Completing a flowchart proof (SE p. 115)</td></tr>
<tr><td>2.7.2 Exploration (20 min.)</td><td>Connecting conditionals to formal proof writing (SE p. 112)</td><td>2.7.4 Guided Practice (10 min.)</td><td>Completing a two-column proof (SE p. 115)</td></tr>
<tr><td>2.7.5 Wrap-Up (~5 min.)</td><td colspan="3">Four quadrant self-reflection on the interest and understanding of this lesson (SE p. 116)</td></tr>
<tr><td rowspan="2">Resources</td><td>Glossary</td><td>Materials</td><td colspan="2">Additional Preparation</td></tr>
<tr><td><u>Key Terms:</u>
Paragraph Proof
Two-column Proof
Definition of Congruence
Flowchart Proof
<u>Additional Glossary:</u> N/A</td><td>N/A</td><td colspan="2">This lesson is the first of a two-lesson series on proofs. Be sure to work through both lessons prior to beginning this one so that you know where the series will end. This can help with inadvertently adding too much information in this lesson by itself.
Consider creating an anchor chart to capture the three different proof formats for the same proof. This will help students get more comfortable with each format as well as with the steps in the process.</td></tr>
</table>

<table>
<tr><td rowspan="4">Teacher Tips</td><td>Common Misconceptions</td><td>Things to Consider</td></tr>
<tr><td>Students may forget to use a given statement as part of their original proof.
Students may skip a step in the proof process.
Students may get lost in the flow of a flowchart proof.
Students may make assumptions based on the diagram instead of using appropriate reasons.</td><td>If one has not already been started, consider having the students create a proof resource booklet, notebook, or type of study guide that will continue to be added to throughout the course. This proof reference would include definitions, postulates, and theorems as well as possible hints (like typical next steps in proofs after a given statement).
Consider referring students back to their experience with the Uno card proofs and the logical flow of the steps.
For 2.7.2 Exploration consider whether your students need a more guided instruction. Take care that your decision does not remove productive struggle from your students or remove the opportunity for the student to engage in discussions with their peers. One possible consideration is to allow students to work independently for question 1 parts a – c and then work with their partner for question part d and beyond. Consider stopping the activity at each vocabulary term (such as paragraph proof) to review with the students and then release them to continue that section. For each section, review their answers to ensure they are on the right track with the outcomes of this activity.</td></tr>
<tr><td>Literacy and Language Routines</td><td>Mathematical Thinking and Reasoning (MTR) Standard</td></tr>
<tr><td>Think-Pair-Share
In 2.7.2 Exploration, students will work together to explore the concept of proofs. They will be set up to work on their own at first to have proper think time and then work with a partner to discuss their answers and move forward with additional exploration together.</td><td>MA.912.MTR.4.1: Discussion and Reflection
The purpose of writing proofs is to provide a justification for why something is true. The act of writing formal proofs develops a student’s ability to present arguments based on evidence while also explaining methods and processes. Create a culture in which your students are comfortable asking questions to their peers and to you.</td></tr>
<tr><td>Differentiate Instruction</td><td colspan="2">In this lesson, students see the same proof presented in all three forms: two-column, paragraph, and flowchart. Consider using a visual aid, such as an anchor chart, to show the clear connection between the forms so that students can see the similarities.</td></tr>
<tr><td colspan="3">Lesson Notes:</td></tr>
</table>

2.7.1 Warm-Up: Miami Art Deco

Component Overview: *Students will observe Art Deco in the Miami area and answer questions related to symmetry.*

2.7.1 Warm-Up: Miami Art Deco

A style known as Art Deco began in Paris in the mid-1920s. Art Deco buildings are characterized by their geometric shapes and symmetry. Two of the most famous examples of Art Deco architecture are the Empire State Building and Chrysler Building in New York City.

Photos of three locations in Miami that use this style of architectural design are shown.

1. Draw a line of symmetry on each picture.
   Answers will vary. Sample answers on images above.

2. Which depiction of Art Deco do you find most interesting?
   Answers will vary.

This activity is designed to connect art concepts to geometry concepts while highlighting Miami.

To add another element of engagement, consider asking students to find other pictures of art deco and describe their geometric shapes and symmetry.

2.7.2 Exploration: Proofs

Component Overview: *Students will connect their prior learning from this unit to writing proofs.*

2.7.2 Exploration: Proofs

A mathematical proof is an argument that shows something is true. A proof should begin with an assumption or a given statement.

1. A statement is given.

   If two angles are supplementary to the same angle, then they are congruent to each other.

   a. What is the hypothesis?
   Two angles are supplementary to the same angle.

   b. What is the conclusion?
   The two angles are congruent to each other.

   c. Draw a picture for the if-then statement.

The hypothesis is the given, or what is known to be true.

The conclusion will need to be shown, or proven, using deductive reasoning.

   d. Discuss with your partner why inductive reasoning is not appropriate for this context. Summarize your discussion.
   Answers will vary. Inductive reasoning is not appropriate because we are not looking at patterns and observations to draw conclusions. Instead, we are looking at facts and general principles.

Lessons 7 and 8 begin the more formal implementation of proofs. Students employ different types of proofs for the same given and prove. The repetitive nature focuses on learning the format of the different types of proofs. Also, by repeating the same thinking, the student is building memory for some common steps.

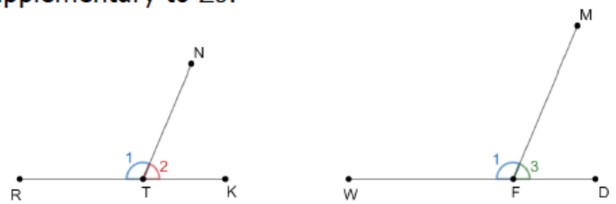
If students are not familiar with the vocabulary word "supplementary." Consider introducing the definition of supplementary angles prior to releasing students to question 1. If students are keeping a proof resource study guide of some type, have the students add this definition.

2.7.2 Exploration: Proofs (continued)

One way to show the deductive reasoning that's necessary to prove the conclusion is a paragraph proof.

A **paragraph proof** is a narrative that gives the steps with justifications that prove a statement as true.

In the diagram, $\angle 1$ is supplementary to $\angle 2$, and $\angle 1$ is supplementary to $\angle 3$.



2. Work with your partner to complete the paragraph proof.

It is given that $\angle 1$ is supplementary to $\angle 2$, and $\angle 1$ is supplementary to $\angle 3$. By definition of supplementary angles $m\angle 1 + m\angle 2 = 180°$ and $m\angle 1 + m\angle 3 = 180°$.

Since $m\angle 1 + m\angle 2 = 180°$ and $180° = m\angle 1 + m\angle 3$, then $m\angle 1 + m\angle 2 = m\angle 1 + m\angle 3$ by the transitive property.

The result of applying the subtraction property of equality is $m\angle 2 = m\angle 3$ By definition of congruence, $\angle 2 \cong \angle 3$.

Another type of proof is the two-column proof.

A **two-column proof** consists of two columns of a chronological list of steps, called statements, and justifications, called reasons.

Since each proof in this activity is the same concept but in a different format, some students may benefit from stations. Consider utilizing stations, consider having students with the strongest understanding begin with the paragraph proof. Students who are struggling should start with the two-column proof. If working with students who are struggling, consider using the two-column proof as a station where students work with you. Utilize the intended flexibility of this lesson to best suit the needs of your students. For students who have started at a proof other than the two-column proof utilize the time they spend with you to verify student answers and to address misconceptions before helping students with the two-column proof.

After completing questions 2 and 3, consider asking the following questions.
What do you think the role is of the given in each of the proofs? Why do you think that it is important to this process?

If students forget the definition of congruence as an option, refer students to Lesson 1.

3. Complete the two-column proof.

Statement	Reason
1. $\angle 1$ is supplementary to $\angle 2$, $\angle 1$ is supplementary to $\angle 3$	1. Given
2. $m\angle 1 + m\angle 2 = 180°$ $m\angle 1 + m\angle 3 = 180°$	2. Definition of *supplementary* angles
3. $m\angle 1 + m\angle 2 = m\angle 1 + m\angle 3$	3. Transitive property
4. $m\angle 2 = m\angle 3$	4. Subtraction property of equality
5. $\angle 2 \cong \angle 3$	5. Definition of congruence

If two figures have equal measurements, then by the **definition of congruence,** the figures are congruent. If congruence is shown through a series of transformations, then the **definition of congruence in terms of rigid motions** was applied.

4. Discuss with your partner why the justification for step 5 is the definition of congruence. Summarize your discussion.
In step 4 we see that $\angle 2$ and $\angle 3$ have equal measures. The definition of congruence states that if two figures have equal measurements, then the figures are congruent. Therefore, we can use that definition as reason $\angle 2$ and $\angle 3$ are congruent.

5. List similarities between the paragraph proof and the two-column proof.
The paragraph proof and two-column proofs both reach the same conclusion, using the same steps and reasoning.

A **flowchart proof** shows the structure of the proof using boxes and connecting arrows. The justifications are written beside or below the box.

Allow students to share out their responses from their partners for other groups to hear. Often, students forget to complete the last step of a proof. Remind students that they know the conclusion of the proof so the steps should lead to that conclusion.

Have students share out their thinking to question 5. Consider leveraging an agree/disagree protocol where students raise their hand if they agree with that choice or disagree with that choice. In both the presentation of the answer and in the agree/disagree component, have students articulate why they chose that response.

2.7.2 Exploration: Proofs (continued)

6. Complete the flowchart proof.

$\angle 1$ is supplementary to $\angle 2$,
$\angle 1$ is supplementary to $\angle 3$

Given

↓

$m\angle 1 + m\angle 2 = 180°$, $m\angle 1 \ + \ m\angle 3 = 180°$

Definition of *supplementary* angles

↓

$m\angle 1 + m\angle 2 = m\angle 1 \ + \ m\angle 3$

Transitive property

↓

$m\angle 2 = m\angle 3$

Subtraction property of equality

↓

$\angle 2 \cong \angle 3$

Definition of congruence

7. Which of the three types of proofs did you find the easiest to follow?
   Answers will vary. I believe the flowchart proof is the easiest to follow.

How does this flowchart proof provide the same structure and information that the paragraph proof did?

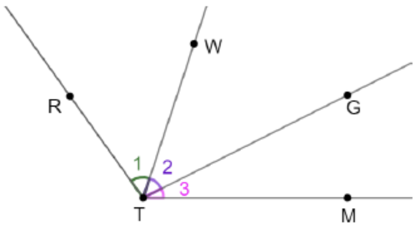
2.7.3 Guided Instruction: Flow Chart Proof

Component Overview: *Teachers will guide students in completing a flowchart proof where the flow from the first statement in not a linear path in nature. Teachers should emphasize that this is a common occurrence when writing proofs.*

2.7.3 Guided Instruction: Flow Chart Proof

In the previous proof, each step depended on the step before it. But this is not always true.

1. Given: $m\angle 1 = m\angle 3$
   Prove: $\angle RTG \cong \angle WTM$



$m\angle 1 = m\angle 3$
Given

$m\angle 1 + m\angle 2 = m\angle RTG$
Angle Addition Postulate

$m\angle 2 + m\angle 3 = m\angle WTM$
Angle Addition Postulate

$m\angle 1 + m\angle 2 = m\angle 2 + m\angle 3$
Addition Property of Equality

$m\angle RTG = m\angle WTM$
Substitution Property of Equality

$\angle RTG \cong \angle WTM$
Definition of Congruence

Pause and ask students what the "given" and "prove" statements mean and discuss the figure shown. This is the first proof that they are seeing in this way, and not all students will recognize what this means. It is also helpful to give students a chance to do a first read of the figure so that they can process the information before moving forward with the notation and formalizing of a proof.

This proof does not have a linear flow like the previous proof. Consider asking these questions:

- *How are the arrows different than in the previous flowchart proof?*
- *Why are three arrows drawn to the one box that states $m\angle RTG = m\angle WTM$?*

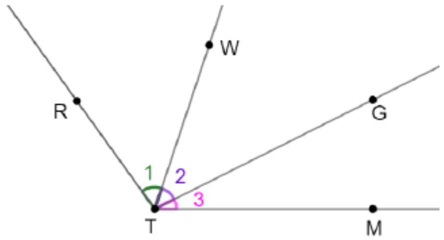
2.7.4 Guided Practice: Two-Column Proof

Component Overview: *Students will complete a two-column proof for the same proof in the previous component. For this activity, students can work independently or with a partner.*

2.7.4 Guided Practice: Two-Column Proof

1. Given: $m\angle 1 = m\angle 3$
   Prove:
   $\angle RTG \cong \angle WTM$

   Complete the two-column proof.



Statement	Reason
1. $m\angle 1 = m\angle 3$	1. Given
2. $m\angle 1 + m\angle 2 = m\angle 2 + m\angle 3$	2. Addition property of equality
3. $m\angle 1 + m\angle 2 = m\angle RTG$	3. Angle addition postulate
4. $m\angle 2 + m\angle 3 = m\angle WTM$	4. Angle addition postulate
5. $m\angle RTG = m\angle WTM$	5. Substitution property of equality
6. $\angle RTG \cong \angle WTM$	6. Definition of congruence

Note that this is the same figure as provided in the 2.7.3 Guided Instruction activity they just completed. Consider releasing students to try this on their own without any additional instruction. Many students may not notice these are the same. One of two events will occur:

- *Students will not realize they are the same image and will try writing the proof again from scratch.*
- *Students will realize they are the same image and will practice the formatting of a two-column proof.*

In both instances, students are getting additional practice with proofs in different capacities.

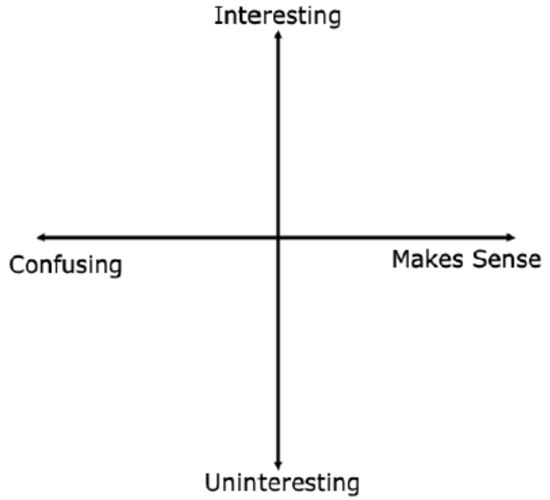
For an additional challenge, have students take this same proof and write it in paragraph proof form.

2.7.5 Wrap-Up: Reflection

Component Overview: *Students will complete a self-reflection on their level of interest and understanding of this lesson. Students should complete this activity independently to provide data for differentiation in the next lesson.*

2.7.5 Wrap-Up: Reflection

1. Plot a point in one of the quadrants below to indicate how you feel about today's lesson.
   Answers will vary.



Since this lesson is the first of a two-part series, it will be important to use the data from student responses in this reflection to structure the release of activities in the next lesson.

2.8 Proofs – Part 2

Lesson Introduction

Benchmarks	MA.912.GR.1.1 Prove relationships and theorems about lines and angles. Solve mathematical and real-world problems involving postulates, relationships, and theorems of lines and angles. MA.912.LT.4.10 Judge the validity of arguments and give counterexamples to disprove statements.	Learning Targets	• I can reflect on the mathematical thinking found in a proof. • I can reason about proofs using deductive reasoning, definitions, and postulates.

Benchmark Coherence	Building On	Working Toward
	MA.8.GR.1.4 Solve mathematical problems involving the relationships between supplementary, complementary, vertical or adjacent angles.	N/A

Essential Questions	• What are the benefits of reflecting on the mathematical thinking found in a proof? • How can I use deductive reasoning, definitions, and postulates to reason about proofs?
Lesson Narrative	In this lesson, students will be moving into the second lesson of a two-lesson series on proofs. Students will be completing proofs from start to finish. They will observe and then participate in the process of brainstorming before writing a proof. Students will complete a flowchart proof and then translate that proof into a two-column proof format. This lesson focuses on the use of MA.K12.MTR.4.1 and is focused more on the learning targets than the actual content of the benchmark. It is very important for students to understand these ideas to master the content in geometry.

Component Summary	2.8.1 Warm-Up (~10 min.)	Bell Work writing a paragraph proof *(SE p. 117)*	2.8.3 Guided Instruction (10 min.)	Brainstorming and writing a flowchart proof *(SE p. 119)*
	2.8.2 Collaboration (15 min.)	Examining brainstorming before writing a proof *(SE p. 117)*	2.8.4 Your Turn! (10 min.)	Translating a flowchart proof to a two-column proof *(SE p. 120)*
	2.8.5 Wrap-Up (~5 min.)	Students summarize their learning *(SE p. 121)*		

Resources	Glossary	Materials	Additional Preparation
	<u>Key Terms:</u> N/A <u>Additional Glossary:</u> • Paragraph Proof • Flowchart Proof	N/A	N/A

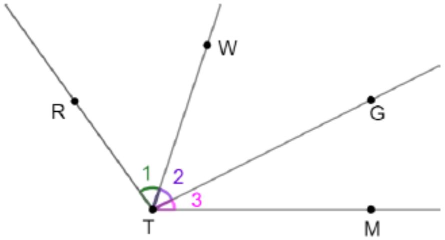
<table>
<tr><td rowspan="4">Teacher Tips</td><td>Common Misconceptions</td><td>Things to Consider</td></tr>
<tr><td>Students may forget to use a given statement as part of their original proof.
Students may skip a step in the proof process.
Students may get lost in the flow of a flowchart proof.
Students may make assumptions instead of using appropriate reasons.</td><td>Consider helping students understand the importance of brain storming before starting a proof. Establish a culture within your class where students can freely share their ideas and to discuss their thinking with their peers and with you.
Consider how allowing students to struggle and problem solve helps them deepen their thinking. Refrain from “rescuing” students too early or rushing students through their productive struggle.
Consider using information from the previous lessons to form a small group of students who are struggling working with proofs. Facilitate these students as they work through 2.8.4 Your Turn!
If time is an issue, then consider having students finish the proof in 2.8.4 Your Turn! for homework.</td></tr>
<tr><td>Literacy and Language Routines</td><td>Mathematical Thinking and Reasoning (MTR) Standard</td></tr>
<tr><td>Clarify, Critique, Correct
2.8.2 Collaboration has students work together to examine the process of brainstorming before writing a proof. After reviewing two examples of brainstorming notes, students can engage in this protocol to clarify, critique, and correct the notes as part of the activity.</td><td>MA.K12.MTR.4.1: Discussion and Reflection
2.8.2 Collaboration is an opportunity for students to continue exploring proofs. Students will engage in collaborative discussions about proofs and brainstorming. They will be expected to analyze the mathematical thinking of others and communicate mathematical ideas, vocabulary, and methods effectively.</td></tr>
<tr><td>Differentiate Instruction</td><td colspan="2">This lesson will be the first experience students have with writing proofs from scratch. Consider providing students with a word bank to choose from that they could match to the corresponding sections of the proofs.</td></tr>
<tr><td colspan="3">Lesson Notes:</td></tr>
</table>

2.8.1 Warm-Up: Bell Work

Component Overview: *Students will write a paragraph proof from the flowchart proof and two-column proof from the previous lesson. Students should complete this activity independently*

2.8.1 Warm-Up: Bell Work

Given: $m\angle 1 = m\angle 3$
Prove: $\angle RTG \cong \angle WTM$



1. Use the flowchart proof and the two-column proof from the previous lesson to write a paragraph proof.
   It is given that $m\angle 1 = m\angle 3$. Using the addition property of equality, we know that $m\angle 1 + m\angle 2 = m\angle 2 + m\angle 3$. Using the angle addition postulate we know that $m\angle 1 + m\angle 2 = m\angle RTG$ and that $m\angle 2 + m\angle 3 = m\angle WTM$. Next we apply the substitution property of equality to determine that $m\angle RTG = m\angle WTM$. By the definition of congruence, we reach our conclusion that $m\angle RTG \cong m\angle WTM$.

For this activity, consider reviewing the instructions with the students prior to having them work on this question. For timing purposes, it is important that students know that this is the same question that they wrote the flowchart and two-column proof for in the prior lesson. If students begin trying to write the proof from scratch, it may take more time than desired. Allow students to look back to proofs from Lesson 7.

2.8.2 Collaboration: Brainstorming Before Writing a Proof

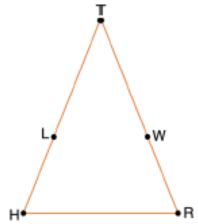
Component Overview: *Students will work collaboratively to examine two examples of brainstorming before writing a proof. For this activity, students should be allowed to work with a partner or small group.*

2.8.2 Collaboration: Brainstorming Before Writing A Proof

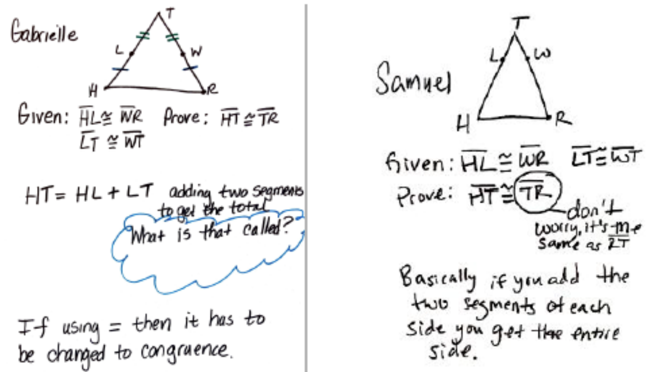
Work with your partner to complete the following.

Gabrielle and Samuel were given the following information.

ΔTHR is shown where $\overline{HL} \cong \overline{WR}$, $\overline{LT} \cong \overline{WT}$.
Prove that $\overline{HT} \cong \overline{TR}$.



Before Gabrielle and Samuel wrote their paragraph proof, they brainstormed. Their notes are shown below.



1. Why did Samuel write "Don't worry it is the same thing as $\overline{RT}$?"
   Samuel wrote "Don't worry it is the same thing as $\overline{RT}$" because $\overline{TR}$ and $\overline{RT}$ represent the same segment between the same two points. When naming a segment, it does not matter which endpoint you name first.

This collaboration is scaffolded to allow students to engage with it with their partner rather than being guided through by the teacher. Teachers can summarize and synthesize what is discovered in the next component. Students may revise their thinking throughout the exploration as they engage with each new reflection. Observe and listen as students work with their partners. Make note of student ideas that you can ask them to share in the whole group 2.8.3 Guided Instruction.

Additional scaffolding questions are provided, if needed, but this should be a primarily student-directed exploration.

For this activity, consider pairing or grouping the students based on their performance in the prior lesson. This activity is directly connected to the concept of writing a proof, which was introduced in the prior lesson.

Consider using high/low combinations based on performance to allow students to work together through the concept of brainstorming before writing a proof while reinforcing the concepts of proofs that were introduced in the prior lesson.

If students have been making a proof resource, encourage students to look through and use their resource to find the names of any definitions, postulates, and theorems.

2.8.2 Collaboration: Brainstorming Before Writing a Proof (continued)

2. Gabrielle made some markings on the triangle she drew. What are her markings meant to show?
The two segments with single dashes are congruent ($\overline{HL} \cong \overline{WR}$), and the two segments with double dashes are congruent ($\overline{LT} \cong \overline{WT}$).

3. Gabrielle wrote "$HT = HL + LT$," but she could not remember what that was called. Samuel wrote "Basically if you add the two segments of each side you get the entire side." What postulate justifies Gabrielle's and Samuel' statements?

   If you and your partner cannot remember the name of the postulate Gabrielle and Samuel are referring to, then look through your workbook to find it.
   The postulate that Gabrielle and Samuel are referring to is the segment addition postulate.

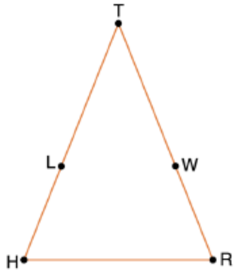
4. Gabrielle only wrote "$HT = HL + LT$," but Samuel mentioned "each side." Write the equation for the other side of the triangle. $RT = RW + WT$

5. Gabrielle wrote "If using = then it has to be changed to congruence." What definition justifies changing from equality to congruence?
The definition of congruence.

While circulating pay particular attention to notations when showing examples of the proof for question 6. Students need to understand the importance of proper notation. For example, measures are added, not the segments themselves, so the segment notations are not used for the equations but are used when stating congruence.

2.8.2 Collaboration: Brainstorming Before Writing a Proof (continued)

6. Apply the brainstorming of Gabrielle and Samuel to finish the paragraph proof. The first sentence has been written for you.



Given: $\overline{HL} \cong \overline{WR}$, $\overline{LT} \cong \overline{WT}$

Prove: $\overline{HT} \cong \overline{TR}$

It is given that $\overline{HL} \cong \overline{WR}$ and $\overline{LT} \cong \overline{WT}$. *By definition of congruence, HL=WR and LT=WT. HL+LT = HT and WR+WT = RT by segment addition postulate. Since HL=WR and LT=WT, then by substitution property of equality WR+WT = HT. Therefore, by the transitive property of equality HT=RT. In conclusion* $\overline{HT} \cong \overline{RT}$ *by the definition of congruence.*

7. Describe the most difficult part of writing the paragraph proof.
   Answers will vary.
8. Describe how Gabrielle and Samuel's brainstorming was helpful.
   Answers will vary.

9. Check your proof with another group and if necessary, resolve any differences.

For question 8, consider having student pairs or groups check with at least two other groups and have each group review and initial that they agree with the group findings. This will allow for multiple perspectives and physical movement in this activity.

2.8.3 Guided Instruction: Flowchart Proof

Component Overview: *Students will first brainstorm and then write a flowchart proof. For this activity, after allowing students to try question 1, have groups share their brainstorming prior to writing the flowchart proof to check for understanding. Then guide students through question 2.*

2.8.3 Guided Instruction: Flowchart Proof

$\overline{AX}$ and $\overline{DY}$ are shown.

Given: K is the midpoint of $\overline{AX}$, L is the midpoint of $\overline{DY}$, and $\overline{AX} \cong \overline{DY}$

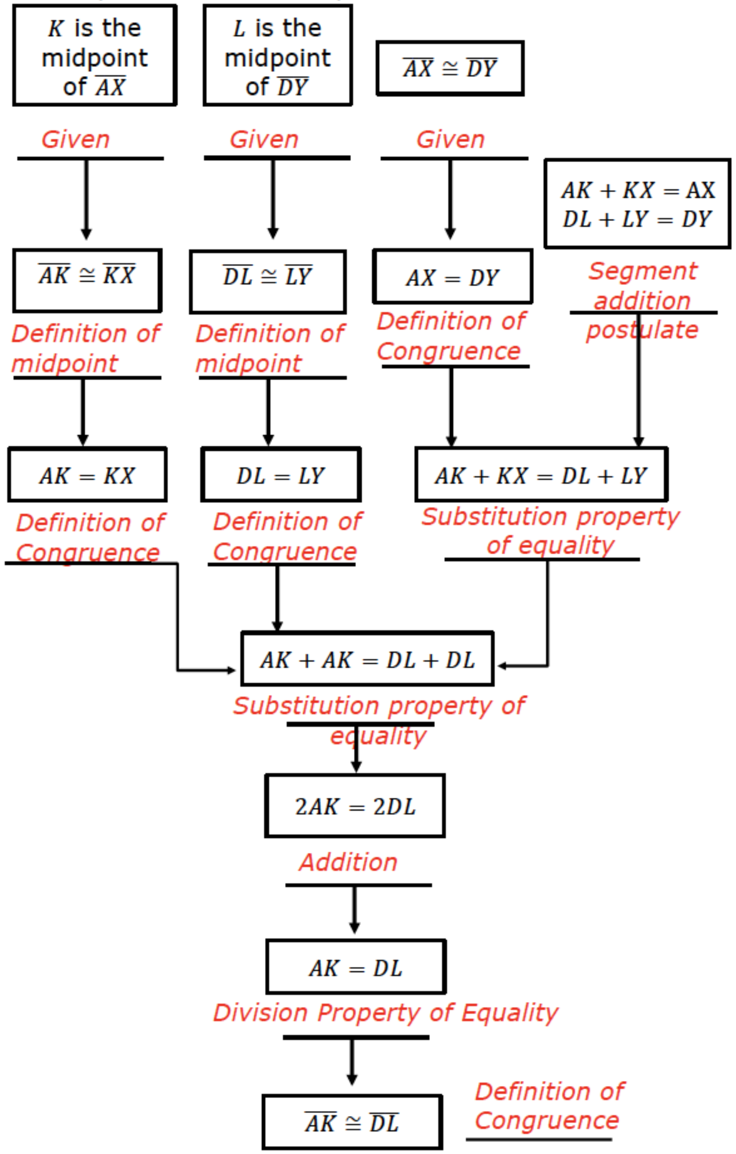
Prove: $\overline{AK} \cong \overline{DL}$

1. Brainstorm with your partner about how to write the proof. Then, write your brainstormed ideas in the space provided. Include any possible definitions or postulates that may need to be used to justify the steps.
   Answers will vary. We discussed needing the definition of bisect. We might also need the segment addition postulate and the definition of congruence.

This proof has a higher cognitive demand. Consider monitoring the length of time that students struggle and help support them where needed. Students may have the misconception that all proofs are direct, which is not the case.

2.8.3 Guided Instruction: Flowchart Proof (continued)

2. Complete the flowchart proof.



Consider creating choice cards as part of this question that students can choose from to complete the boxes if that level of scaffolding is needed.

If students are ready for a challenge, have students close their books and attempt this flowchart proof without the use of a template to see how they approach the problem from start to finish.

2.8.4 Your Turn!

Component Overview: *Students will use the flowchart from the prior activity to complete the two-column proof. For this activity, students should work independently to practice what they've been learning in this lesson.*

2.8.4 Your Turn!

1. Use the previous flowchart to complete the two-column proof.

Statement	Reason
1. K is the midpoint of $\overline{AX}$ and L is the midpoint of $\overline{DY}$	1. Given
2. $\overline{AK} \cong \overline{KX}$ and $\overline{DL} \cong \overline{LY}$	2. *Definition of midpoint*
3. $AK = KX$ and $DL = LY$	3. *Definition of congruence*
4. $\overline{AX} \cong \overline{DY}$	4. Given
5. $AX = DY$	5. Definition of congruence
6. $AK + KX = AX$ and $DL + LY = DY$	6. *Segment addition postulate*
7. $AK + KX = DL + LY$	7. Substitution property of equality
8. $AK + AK = DL + DL$	8. *Substitution property of equality*
9. $2AK = 2DL$	9. *Addition*
10. $AK = DL$	10. *Division property of equality*
11. $\overline{AK} \cong \overline{DL}$	11. *Definition of congruence*

This activity can be modified to be scaffolded based on student needs. For example:

- *If students are understanding the concept very well, consider having them complete the proof without the table in the book from start to finish*
- *If students are struggling with this concept, consider the use of choice cards to put together pieces of the puzzle in this process*
- *If students are doing fairly well, leave the activity unmodified and do it as is.*

2.8.5 Wrap-Up: Cool Down

***Component Overview**: Students write a short summary of their learning in the lesson. Student responses can be used to identify misconceptions and plan for upcoming instruction.*

2.8.5 Wrap-Up: Cool Down

1. Write a summary of what you learned in this lesson.
   Answers will vary.

Consider using student responses from prior self-reflections to consider a student's learning progress. Recording this information and pairing it with assessment data provides insightful feedback for student learning progression.